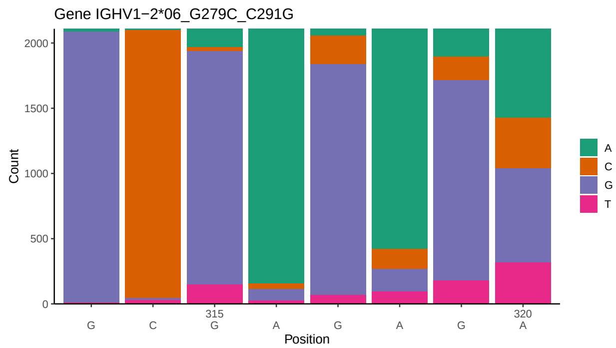
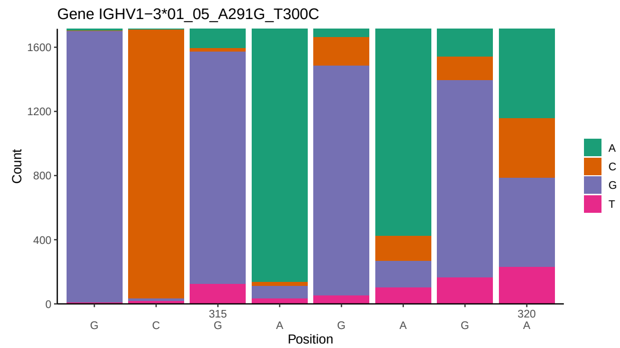
OGRDBstats Report

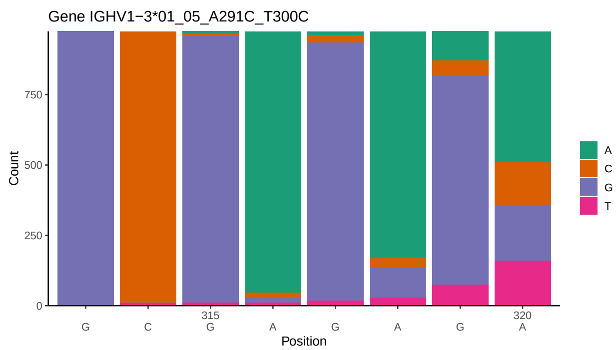
Contents

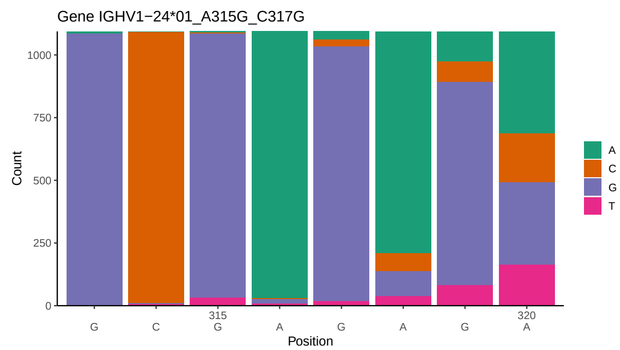
1 Novel sequence analysis

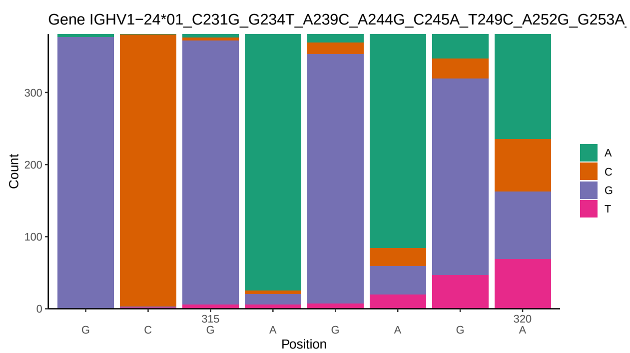
1.1 End-nucleotide composition

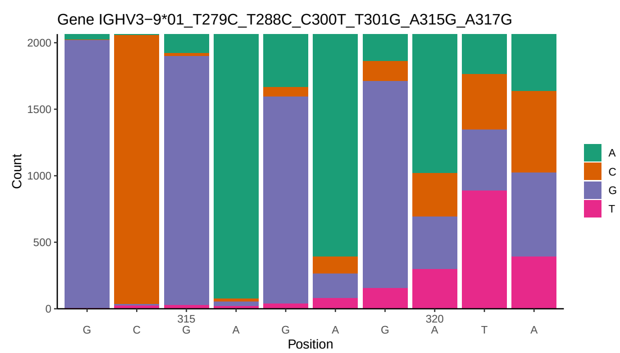


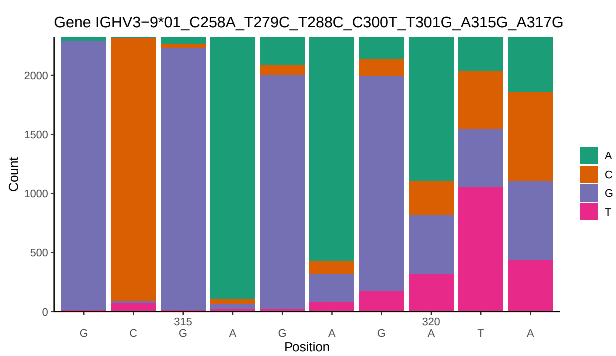


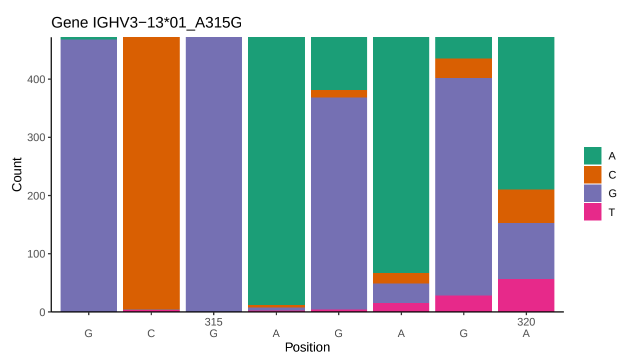


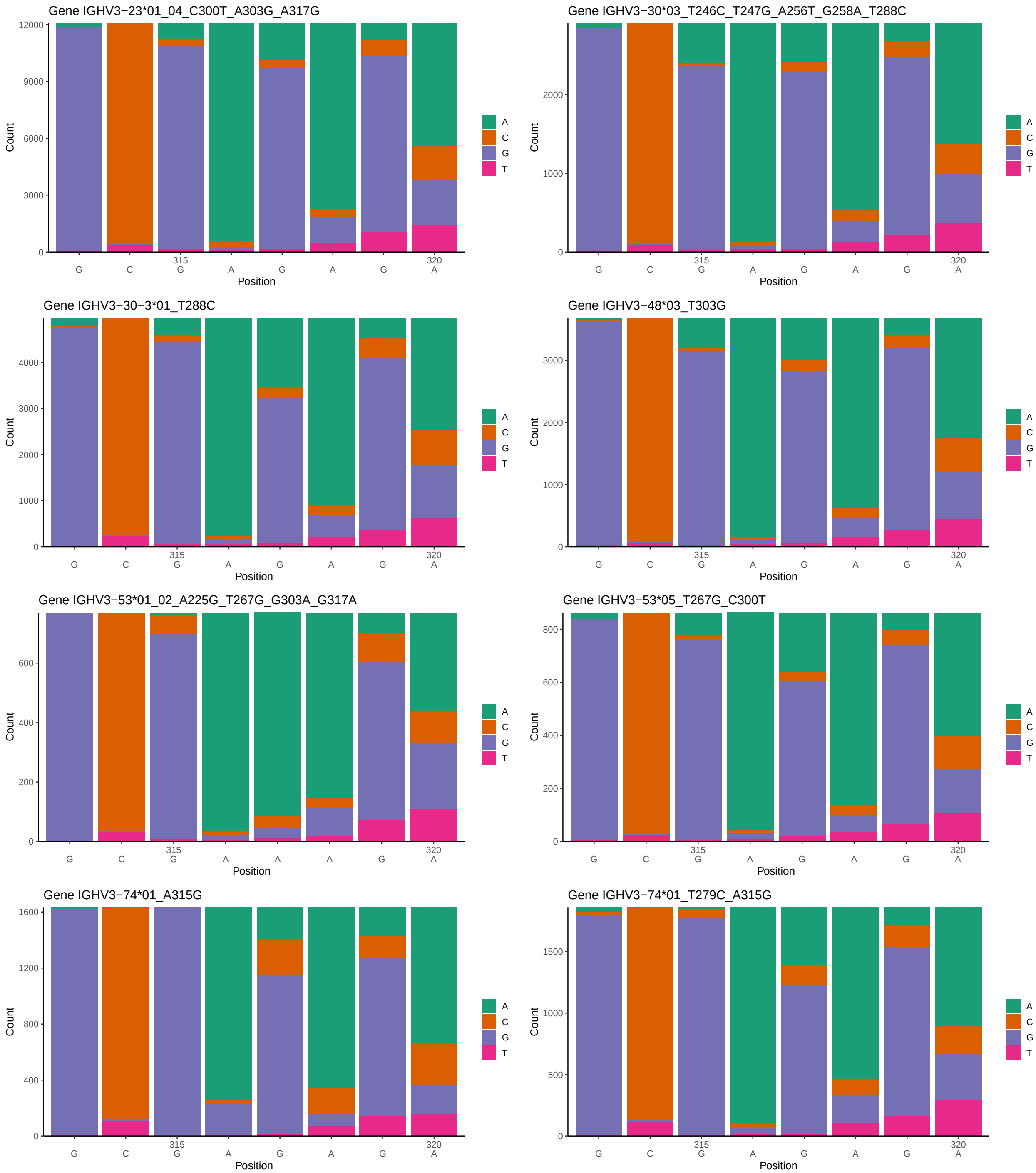




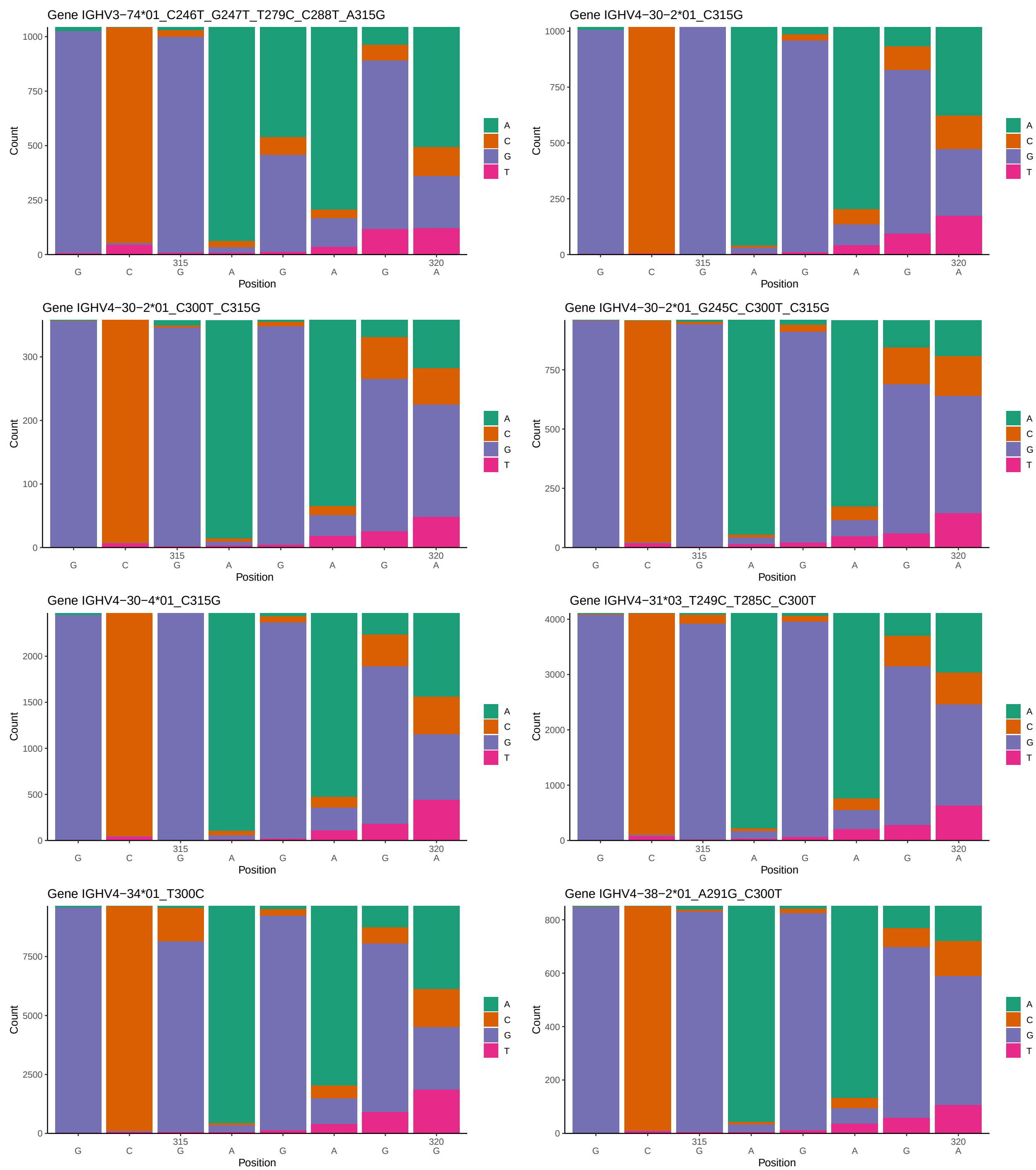








Gene IGHV3−23*01_04_C300T_A303G_A317G
Gene IGHV3−30*03_T246C_T247G_A256T_G258A_T288C
Gene IGHV3−30−3*01_T288C
Gene IGHV3−48*03_T303G
Gene IGHV3−53*01_02_A225G_T267G_G303A_G317A
Gene IGHV3−53*05_T267G_C300T
Gene IGHV3−74*01_A315G
Gene IGHV3−74*01_T279C_A315G
Count
Position
A
C
G
T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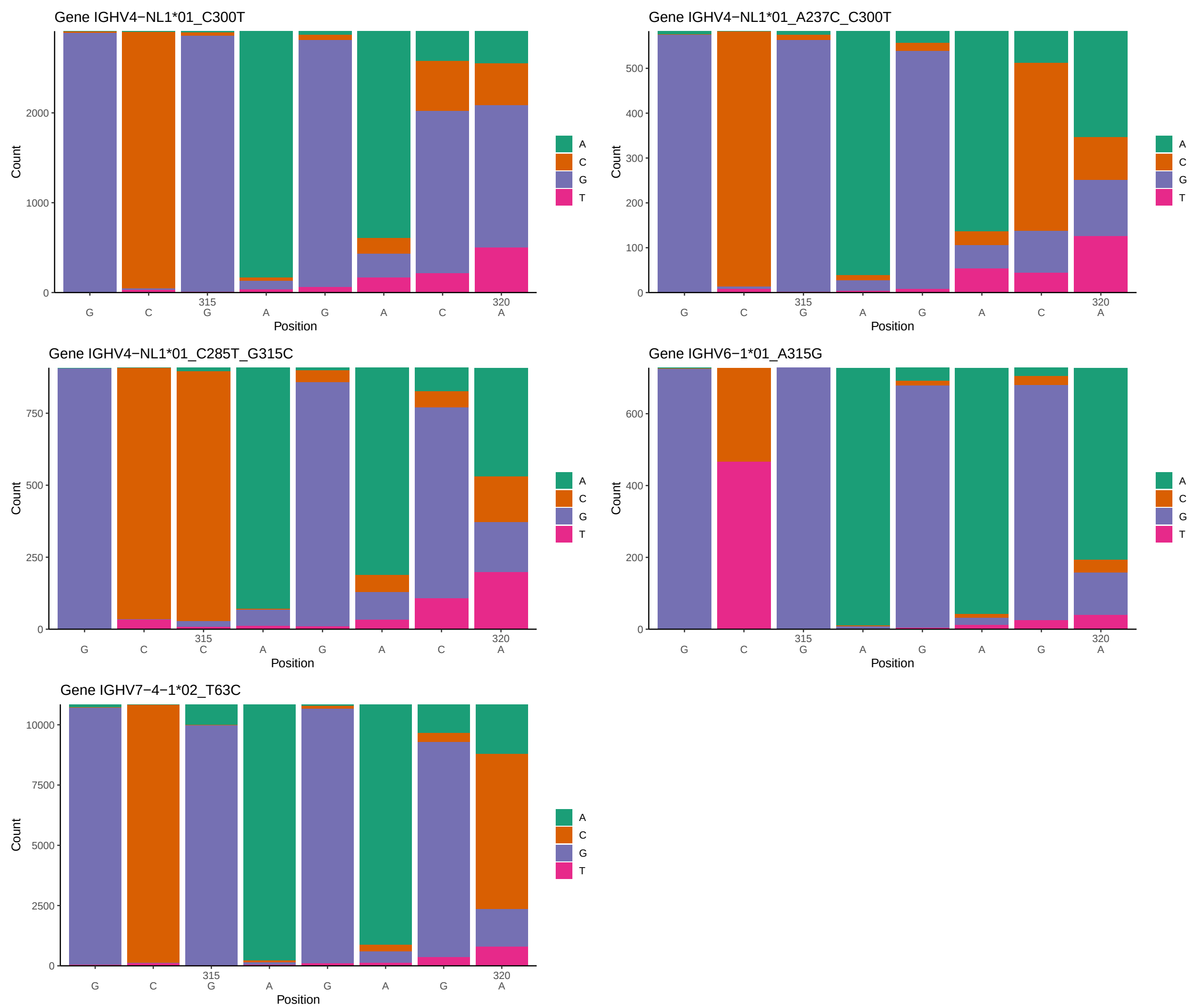
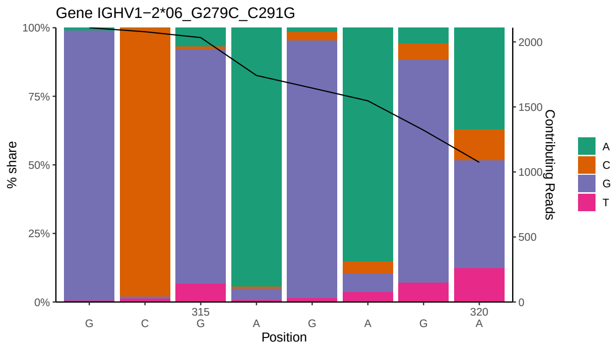
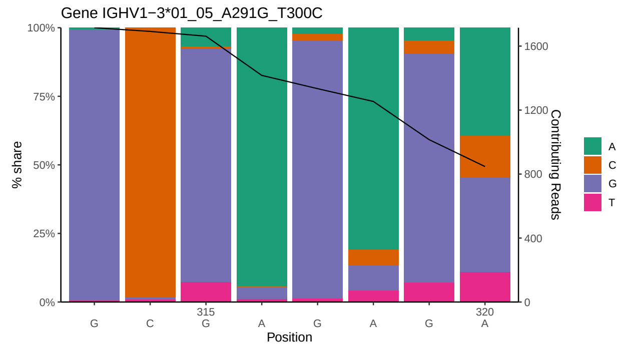
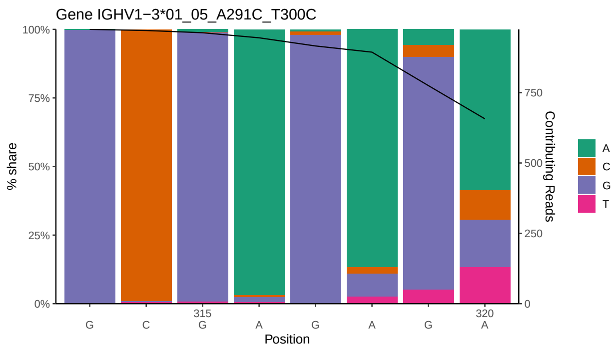
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−2*01_C300T_C315G
Gene IGHV4−30−2*01_G245C_C300T_C315G
Gene IGHV4−30−4*01_C315G
Gene IGHV4−31*03_T249C_T285C_C300T
Gene IGHV4−34*01_T300C
Gene IGHV4−38−2*01_A291G_C300T
Count
Position
A
C
G
T

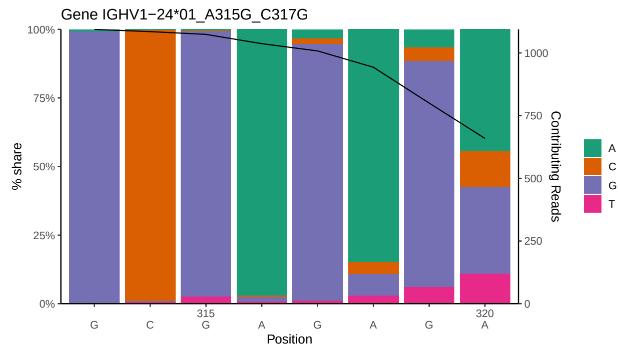

Gene IGHV4−NL1*01_C300T
Gene IGHV4−NL1*01_A237C_C300T
Gene IGHV4−NL1*01_C285T_G315C
Gene IGHV6−1*01_A315G
Gene IGHV7−4−1*02_T63C
Count
Position
A
C
G
T
315
320

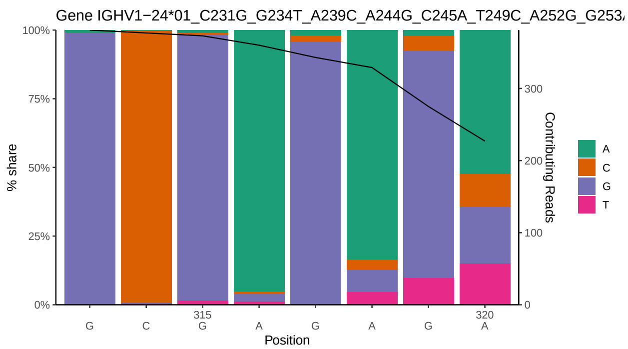
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

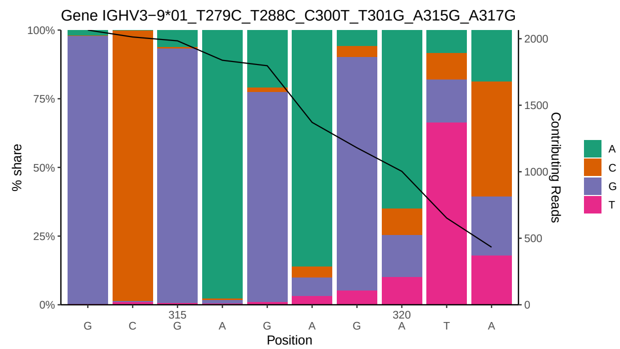


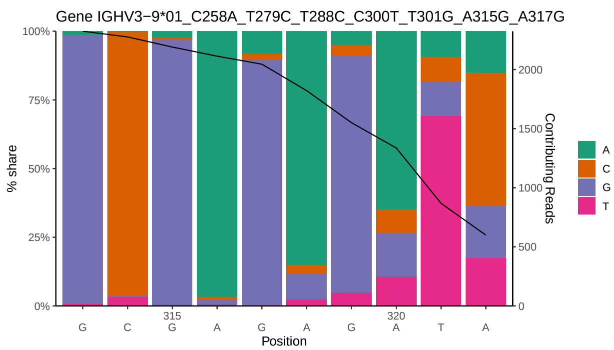


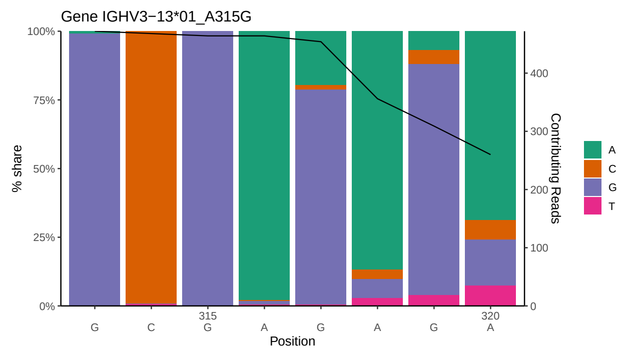


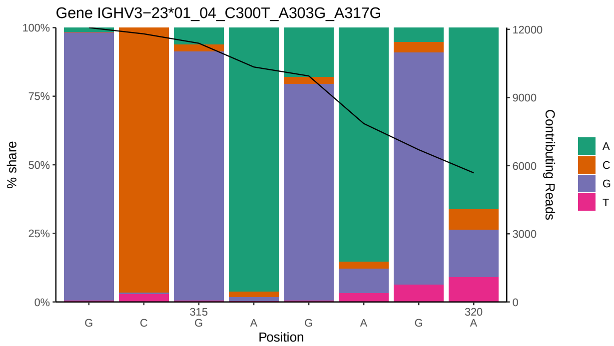








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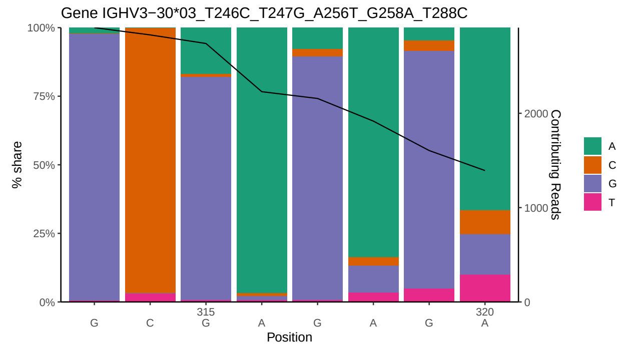
Contributing Reads
Position
A
C
G
T


Gene IGHV3−30*03_T246C_T247G_A256T_G258A_T288C
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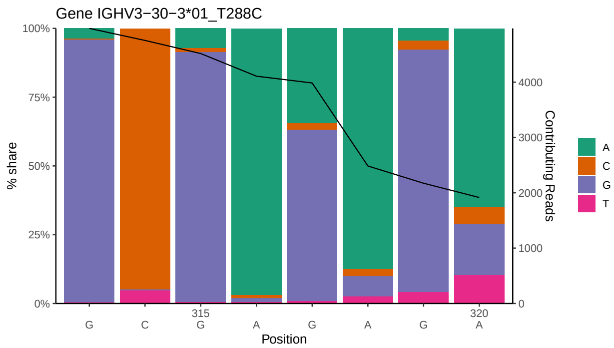
Contributing Reads
Position
A
C
G
T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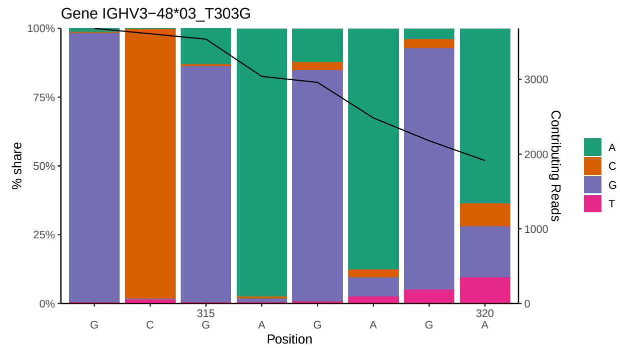
Contributing Reads
Position
A
C
G
T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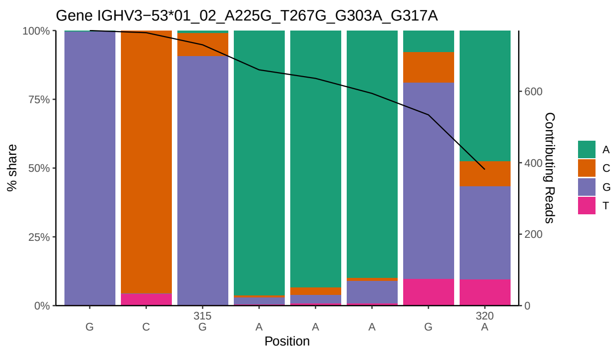
Contributing Reads
Position
A
C
G
T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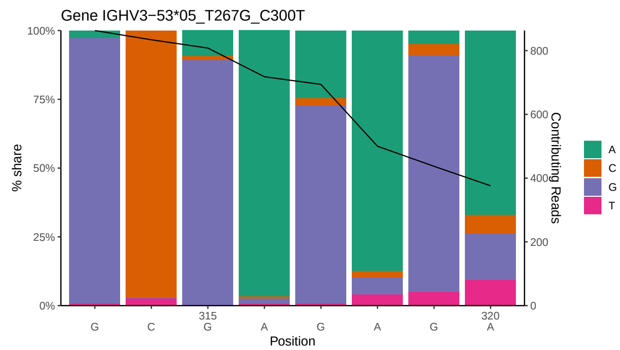
Contributing Reads
Position
A
C
G
T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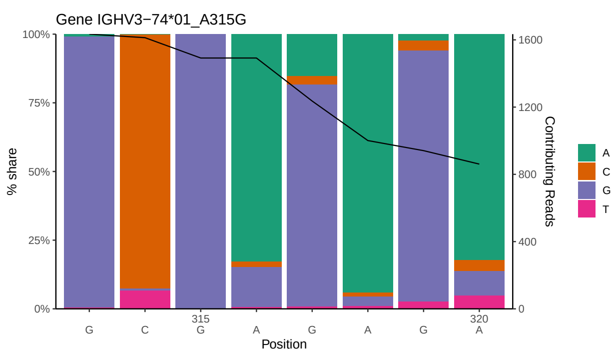
% share
Contributing Reads
Position
A
C
G
T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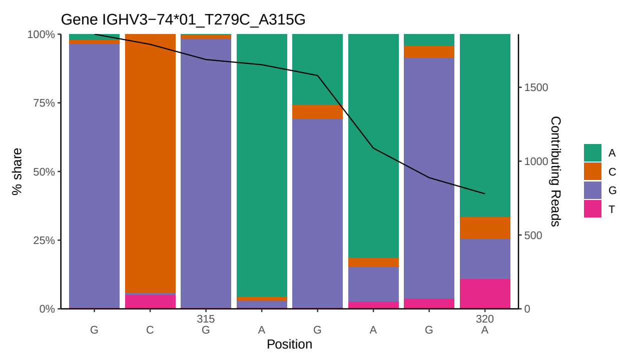
Contributing Reads
Position
A
C
G
T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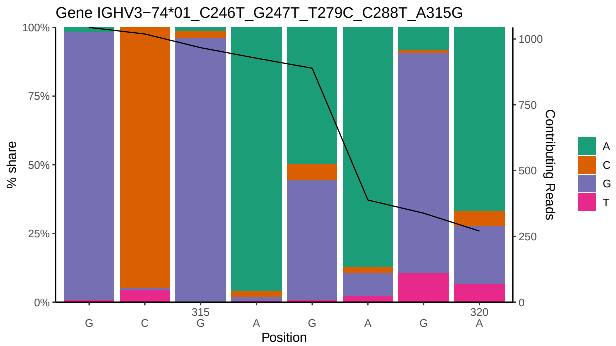
% share
Contributing Reads
Position
A
C
G
T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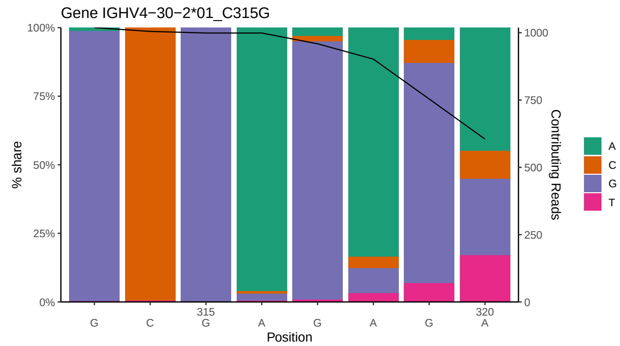
% share
Position
Contributing Reads
A
C
G
T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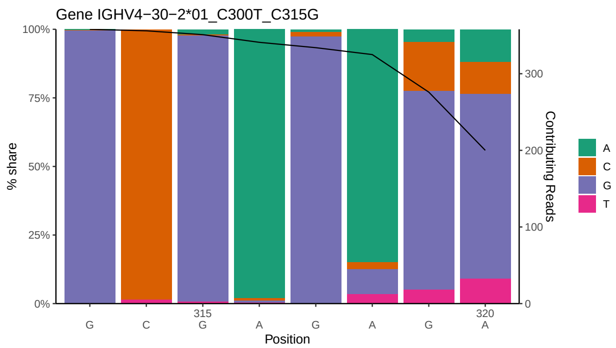
Position
Contributing Reads
A
C
G
T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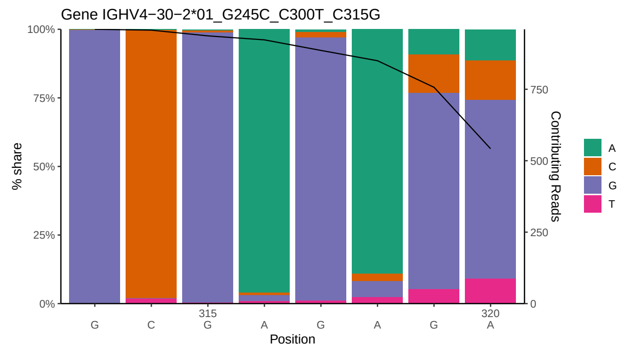
Position
Contributing Reads
A
C
G
T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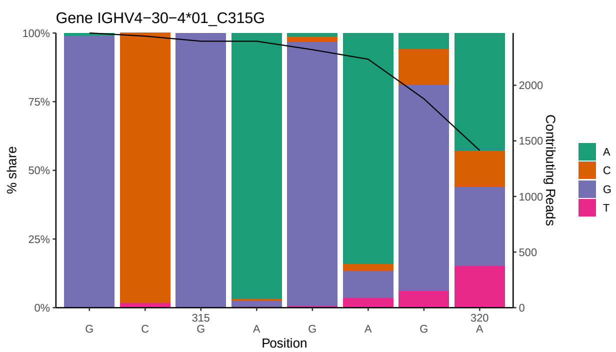
% share
Position
Contributing Reads
A
C
G
T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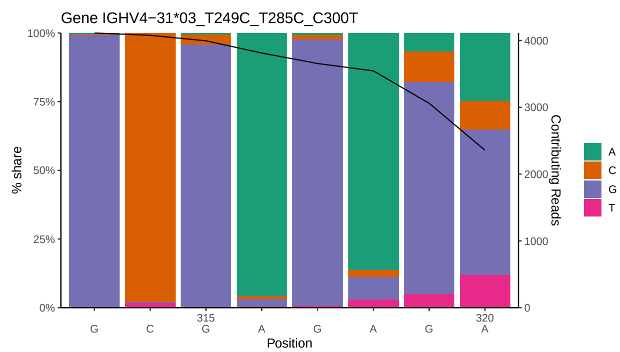
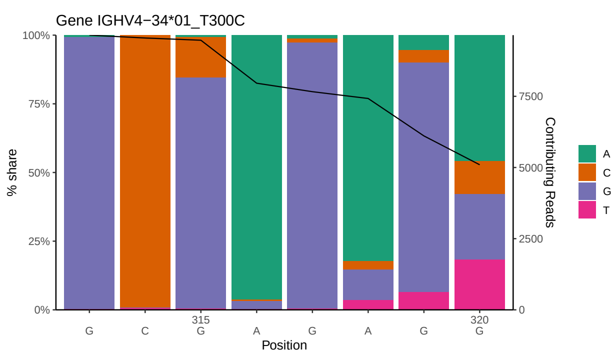
Position
Contributing Reads
A
C
G
T


Gene IGHV4−31*03_T249C_T285C_C300T
% share
Position
Contributing Reads
A
C
G
T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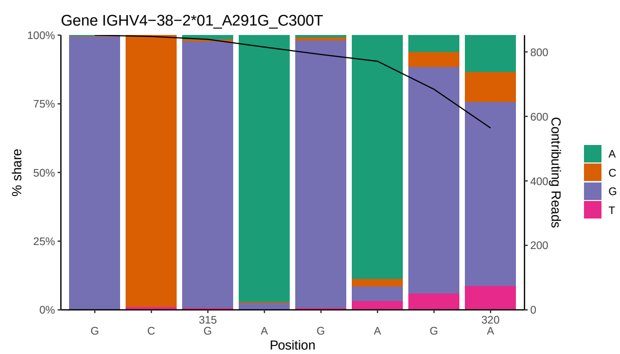
Position
Contributing Reads
A
C
G
T


Gene IGHV4−38−2*01_A291G_C300T
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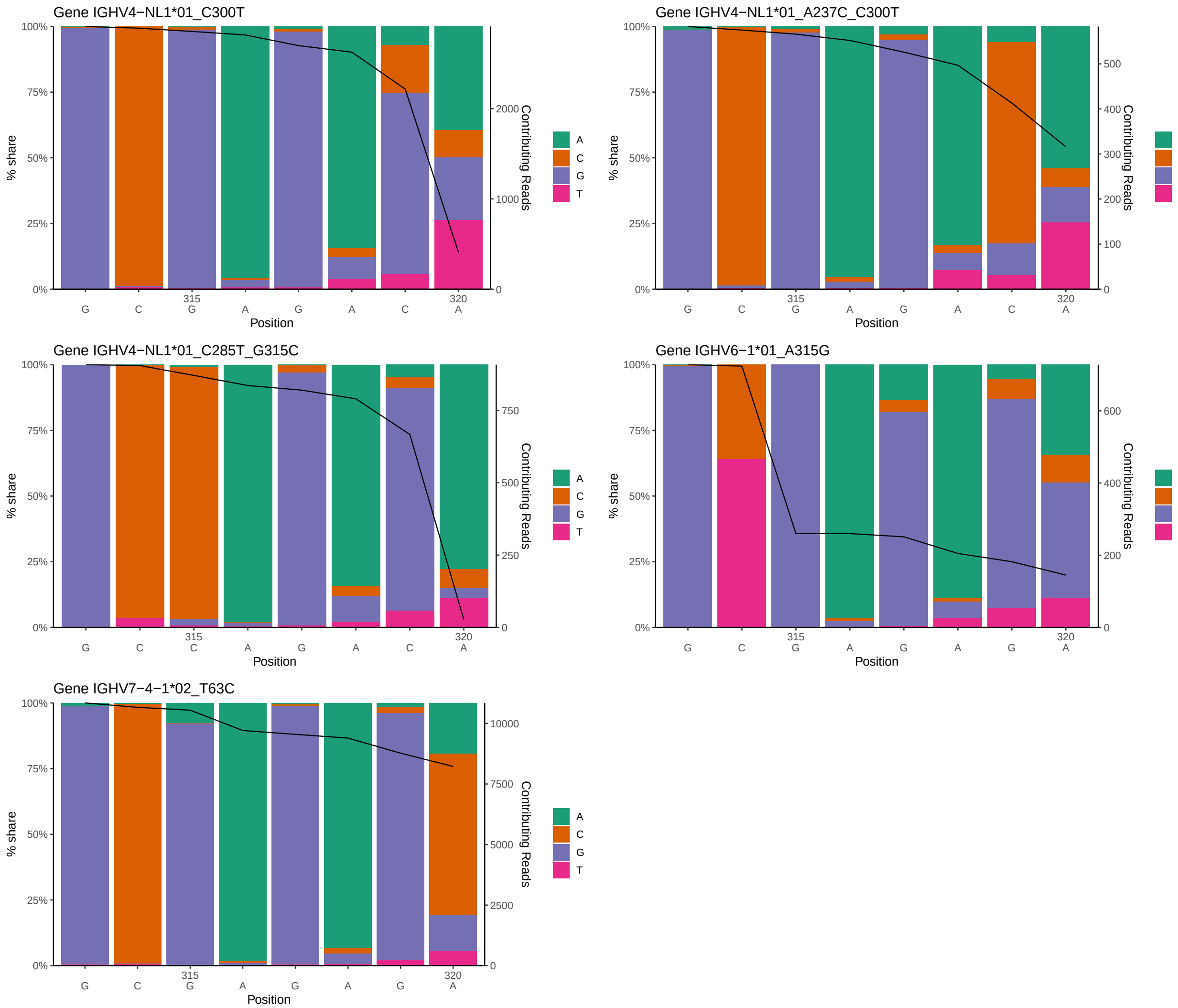
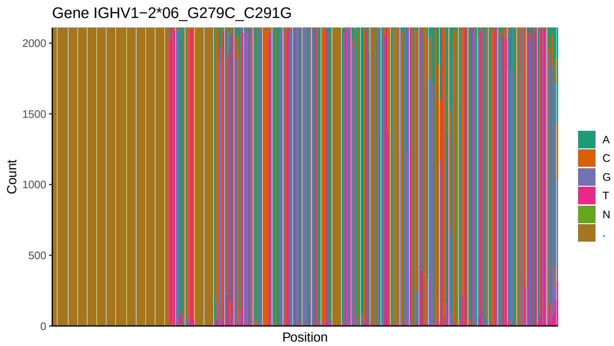
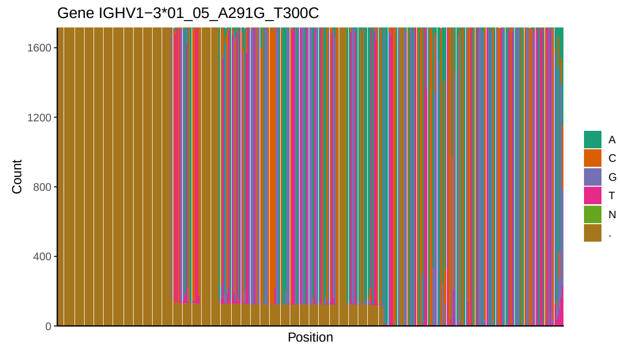
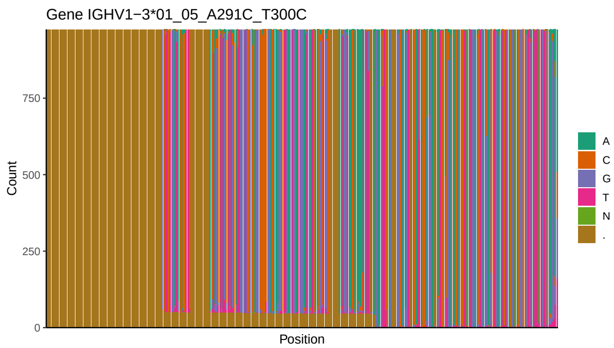
Position
Contributing Reads
A
C
G
T

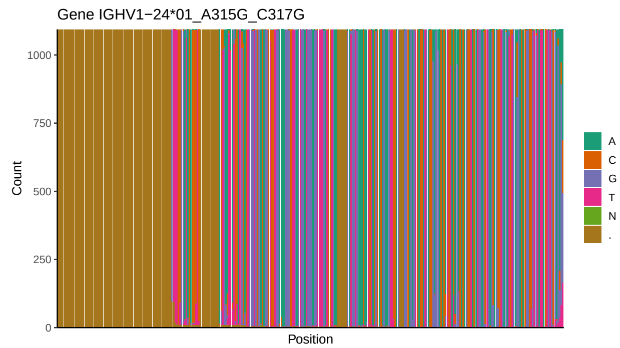

Gene IGHV4−NL1*01_C300T
Gene IGHV4−NL1*01_A237C_C300T
Gene IGHV4−NL1*01_C285T_G315C
Gene IGHV6−1*01_A315G
Gene IGHV7−4−1*02_T63C
% share
Contributing Reads
Position
A
C
G
T

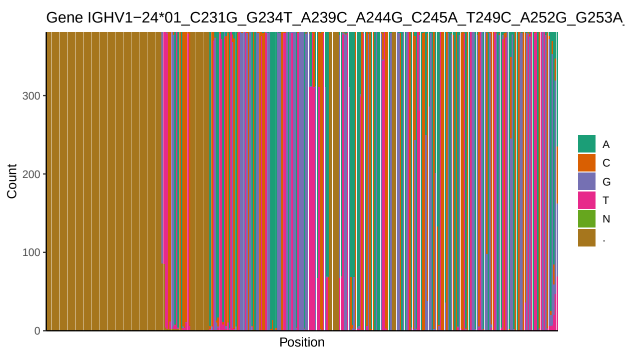
1.3 Whole-sequence composition of each assigned read

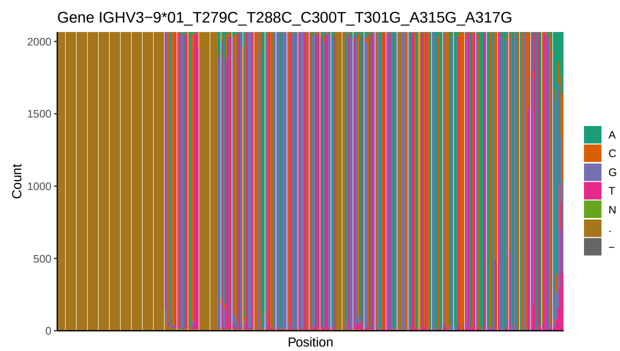


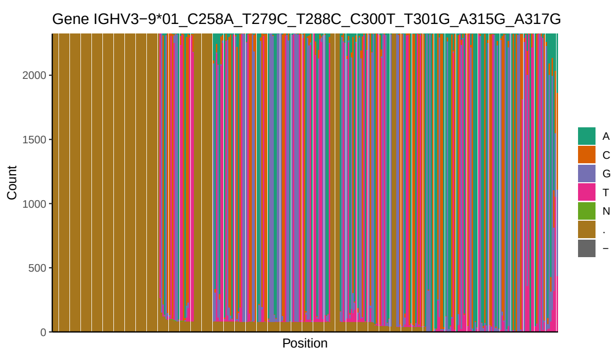


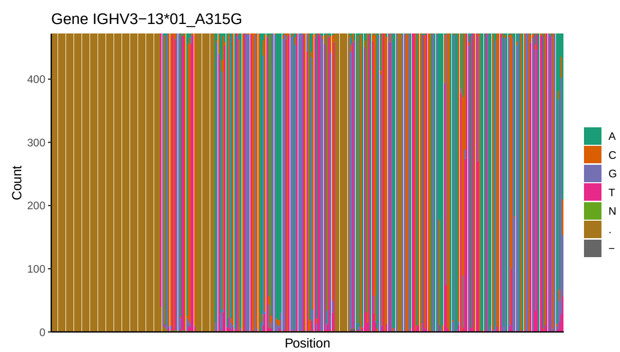


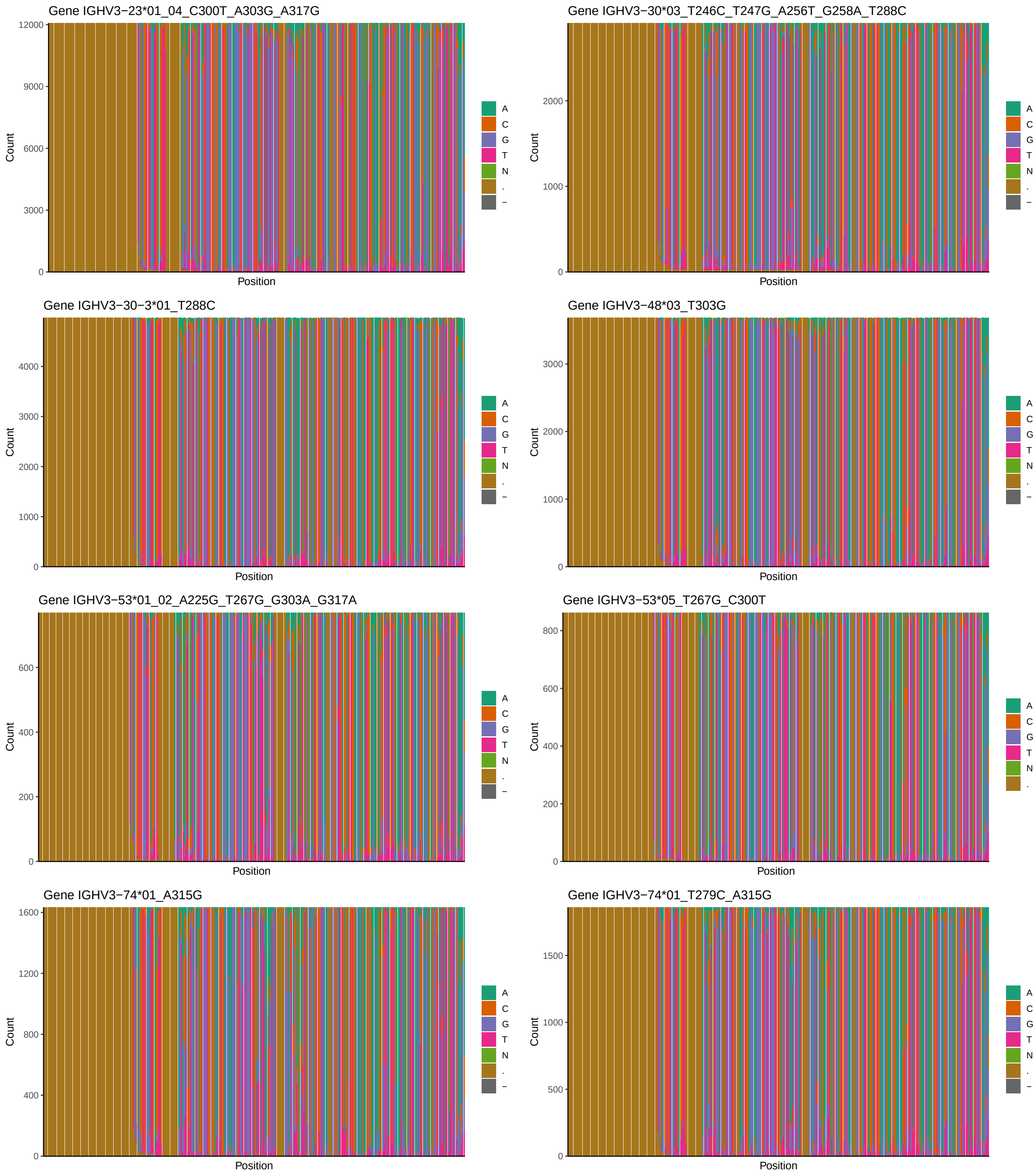


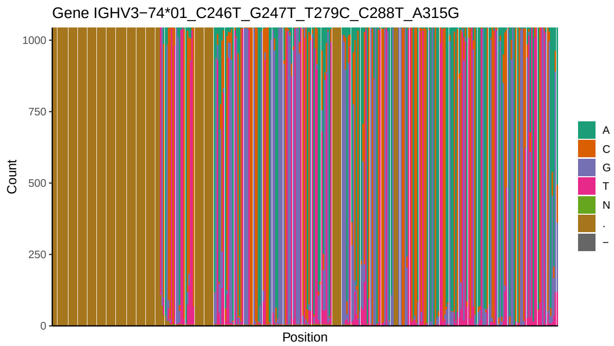









Gene IGHV3−23*01_04_C300T_A303G_A317G
Gene IGHV3−30*03_T246C_T247G_A256T_G258A_T288C
Gene IGHV3−30−3*01_T288C
Gene IGHV3−48*03_T303G
Gene IGHV3−53*01_02_A225G_T267G_G303A_G317A
Gene IGHV3−53*05_T267G_C300T
Gene IGHV3−74*01_A315G
Gene IGHV3−74*01_T279C_A315G
Count
Position
A
C
G
T
N
.
−


Gene IGHV3−74*01_C246T_G247T_T279C_C288T_A315G
Count
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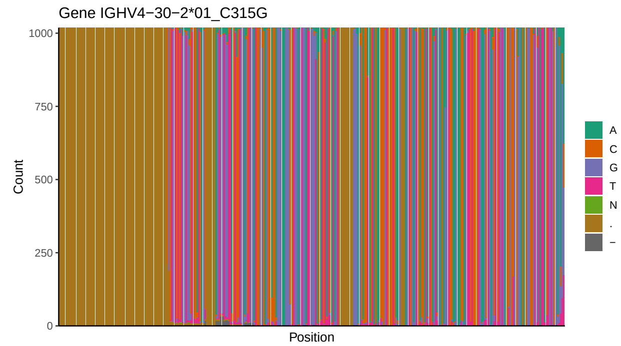
250
500
750
1000
Position
A
C
G
T
N
.
−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Count
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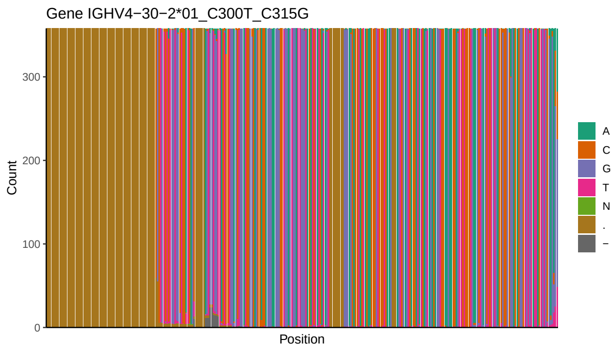
250
500
750
1000
Position
A
C
G
T
N
.
−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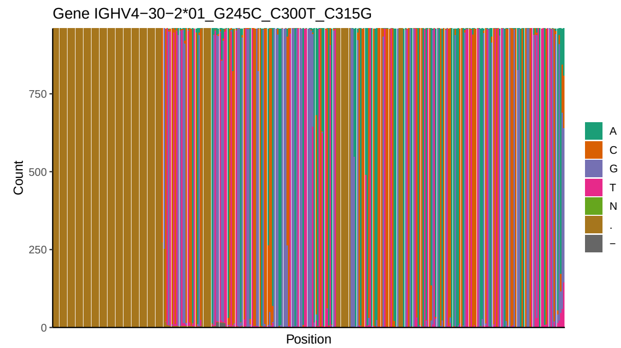
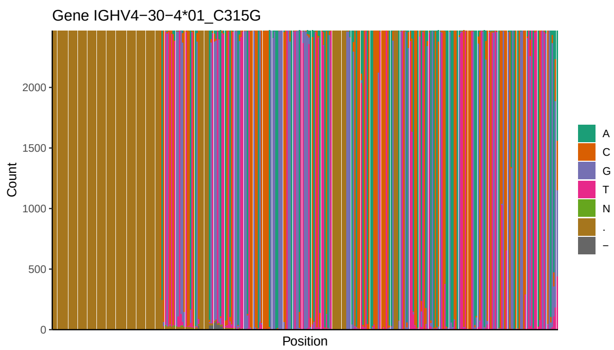
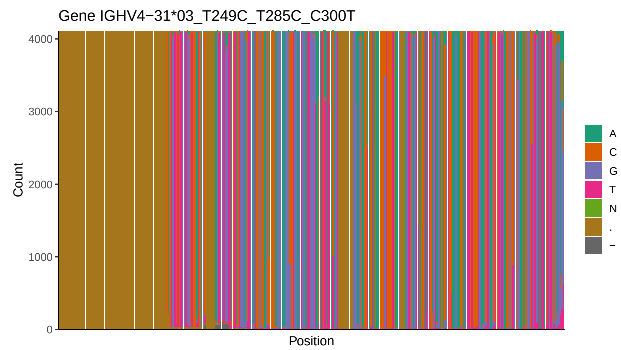
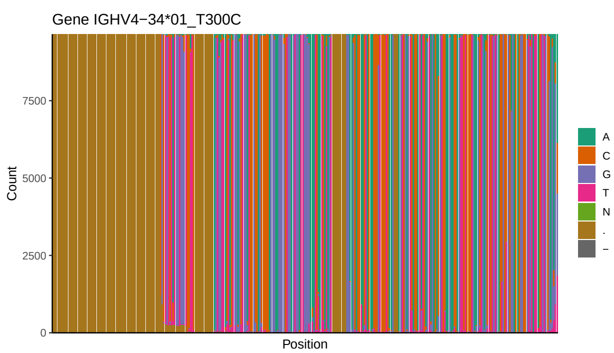
Count
0
100
200
300
Position
A
C
G
T
N
.
−


Gene IGHV4−30−2*01_G245C_C300T_C315G
Count
0
250
500
750
Position
A
C
G
T
N
.
−


Gene IGHV4−30−4*01_C315G
Count
0
500
1000
1500
2000
Position
A
C
G
T
N
.
−


Gene IGHV4−31*03_T249C_T285C_C300T
Count
0
1000
2000
3000
4000
Position
A
C
G
T
N
.
−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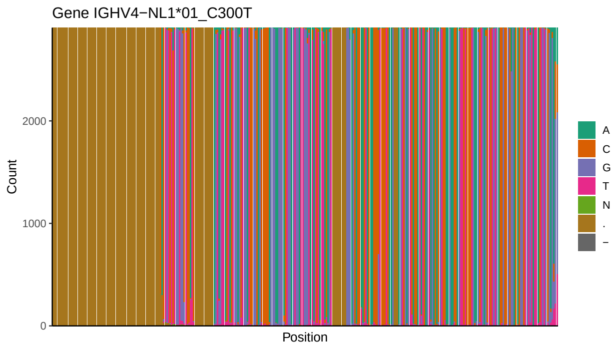
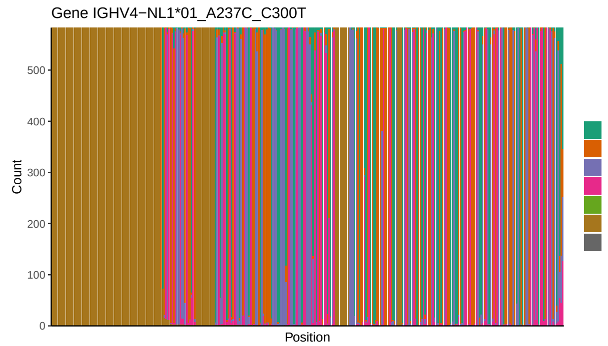
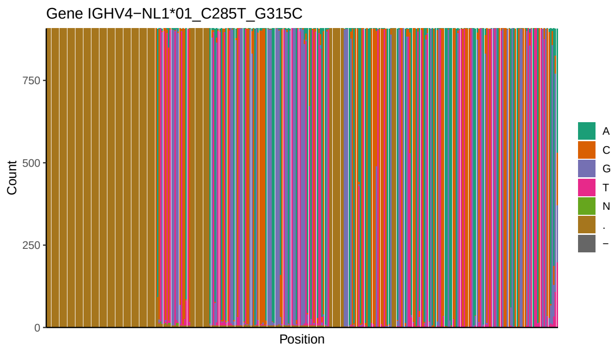
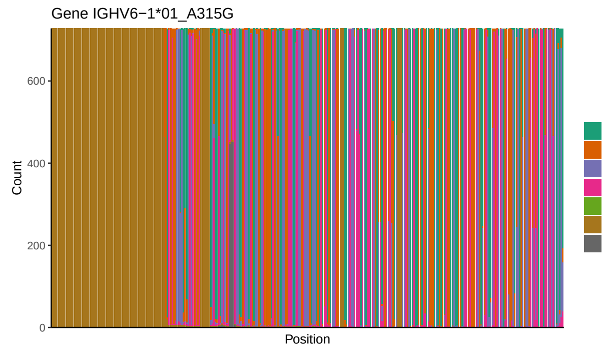
Count
0
2500
5000
7500
Position
A
C
G
T
N
.
−

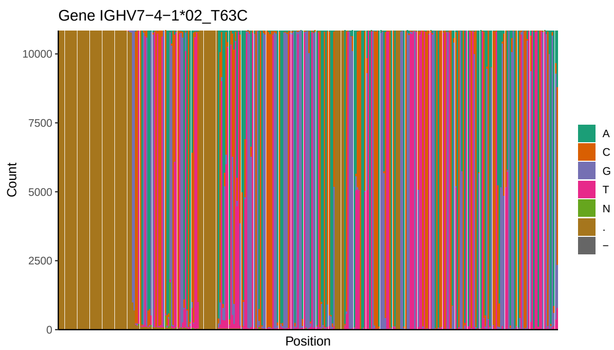

Gene IGHV4−38−2*01_A291G_C300T
Count
0
200
400
600
800
Position
A
C
G
T
N
.
−



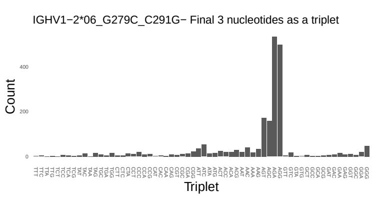


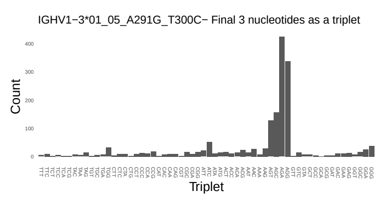


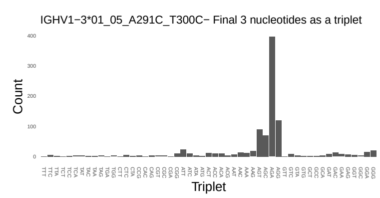


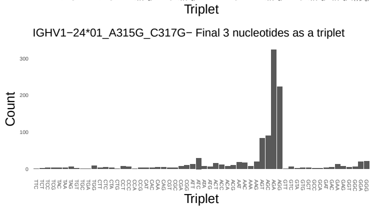


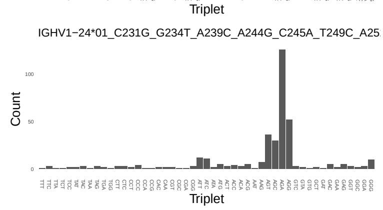
1.4 Final three nucleotides: frequency of each observed triplet

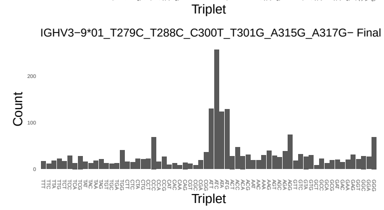


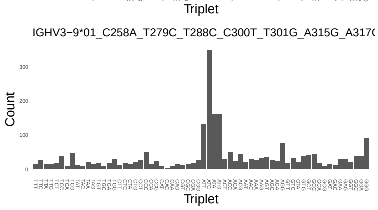


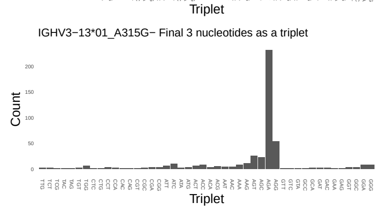


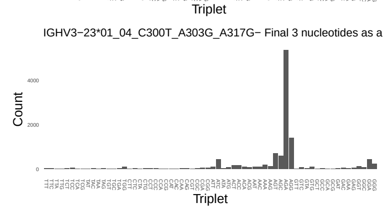


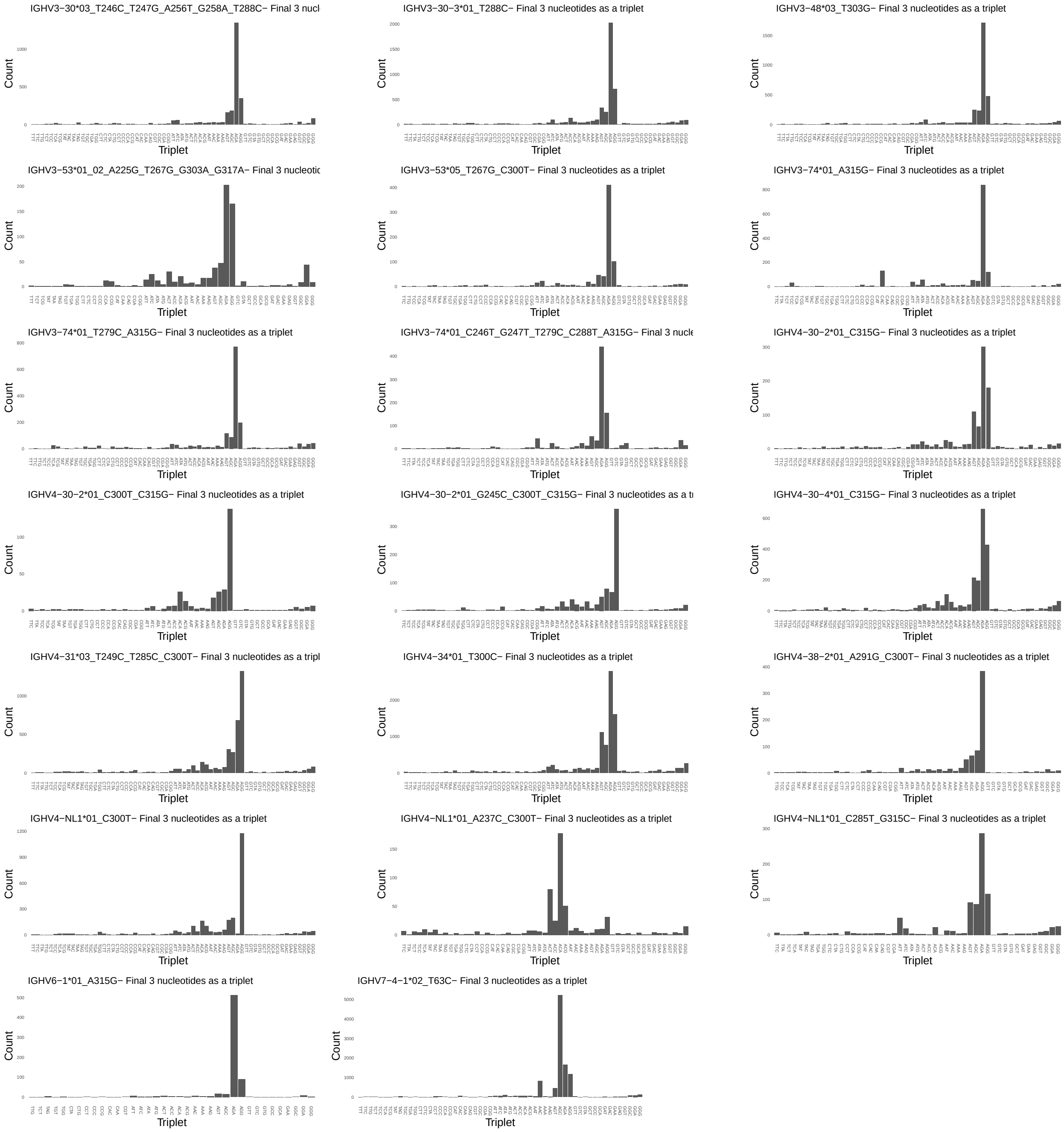




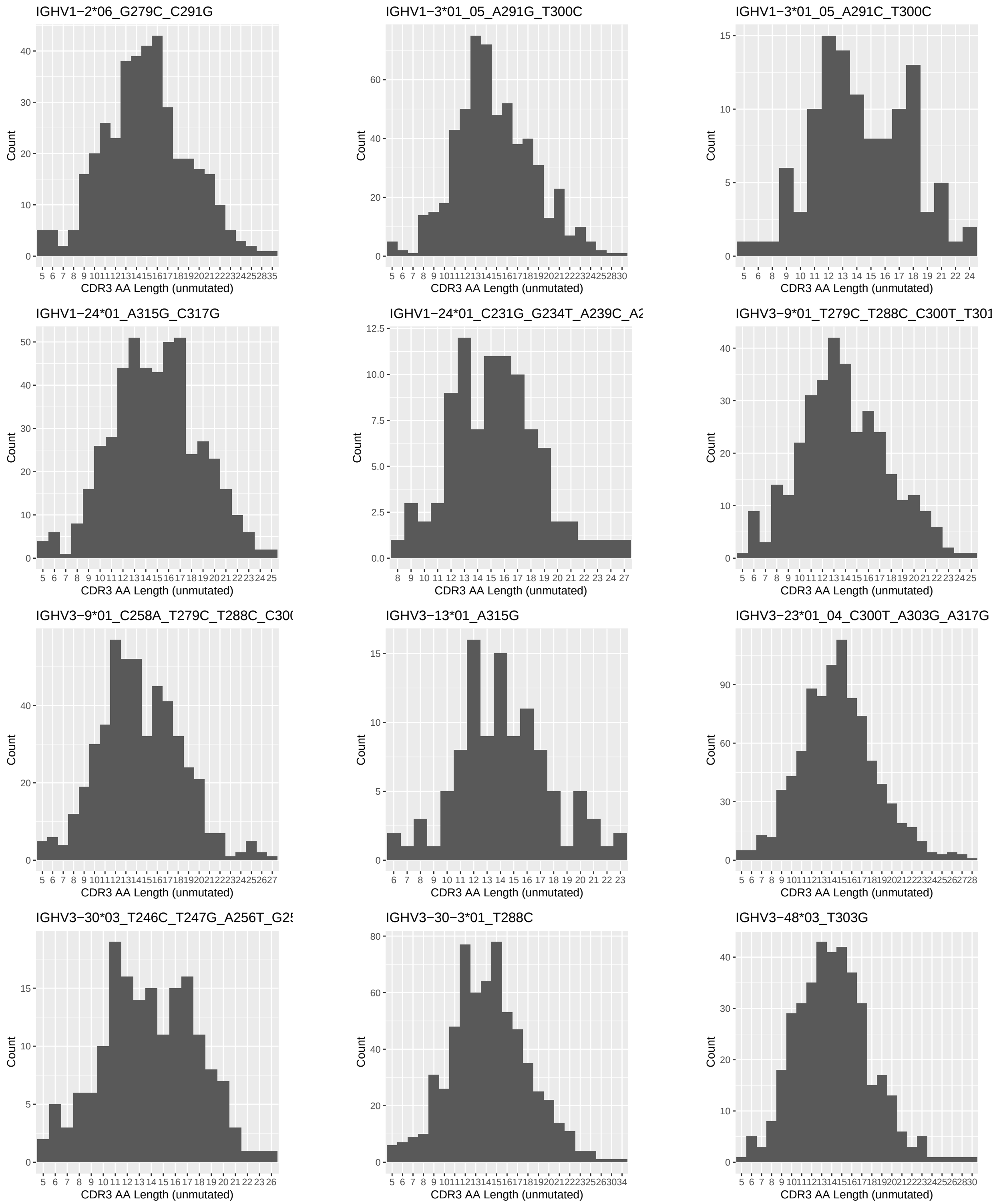


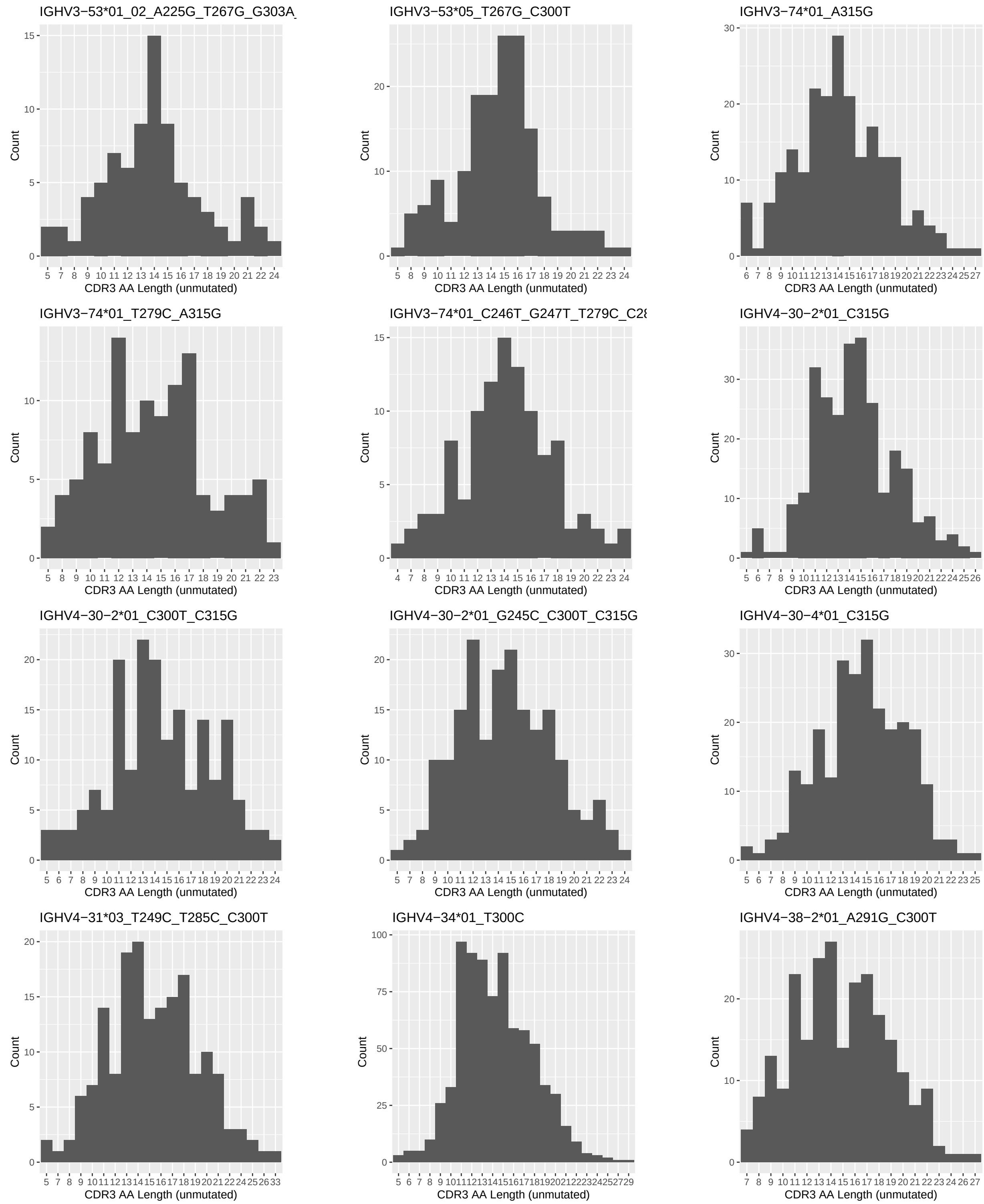


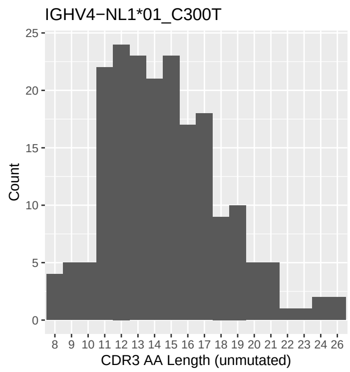


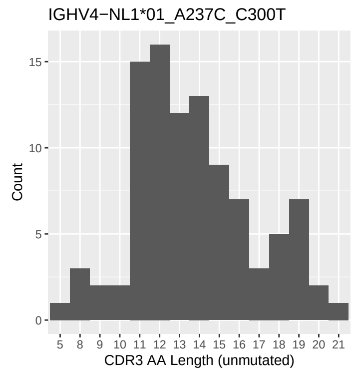

IGHV3−30*03_T246C_T247G_A256T_G258A_T288C− Final 3 nucl
IGHV3−30−3*01_T288C− Final 3 nucleotides as a triplet
IGHV3−48*03_T303G− Final 3 nucleotides as a triplet
IGHV3−53*01_02_A225G_T267G_G303A_G317A− Final 3 nucleotic
IGHV3−53*05_T267G_C300T− Final 3 nucleotides as a triplet
IGHV3−74*01_A315G− Final 3 nucleotides as a triplet
IGHV3−74*01_T279C_A315G− Final 3 nucleotides as a triplet
IGHV3−74*01_C246T_G247T_T279C_C288T_A315G− Final 3 nucle
IGHV4−30−2*01_C315G− Final 3 nucleotides as a triplet
IGHV4−30−2*01_C300T_C315G− Final 3 nucleotides as a triplet
IGHV4−30−2*01_G245C_C300T_C315G− Final 3 nucleotides as a t
IGHV4−30−4*01_C315G− Final 3 nucleotides as a triplet
IGHV4−31*03_T249C_T285C_C300T− Final 3 nucleotides as a tripl
IGHV4−34*01_T300C− Final 3 nucleotides as a triplet
IGHV4−38−2*01_A291G_C300T− Final 3 nucleotides as a triplet
IGHV4−NL1*01_C300T− Final 3 nucleotides as a triplet
IGHV4−NL1*01_A237C_C300T− Final 3 nucleotides as a triplet
IGHV4−NL1*01_C285T_G315C− Final 3 nucleotides as a triplet
IGHV6−1*01_A315G− Final 3 nucleotides as a triplet
IGHV7−4−1*02_T63C− Final 3 nucleotides as a triplet
Count
Triplet

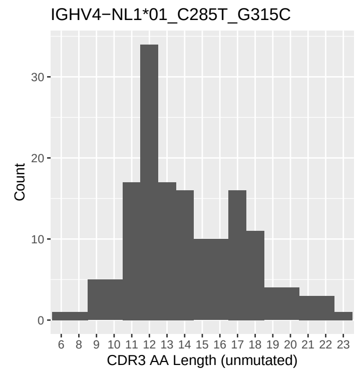
1.5 CDR3 length distribution, in assignments to novel alleles

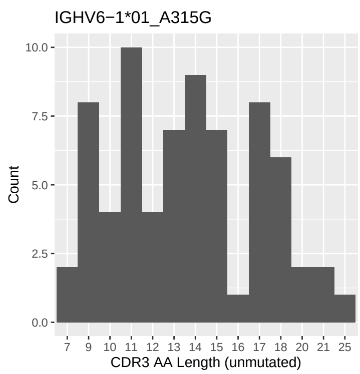


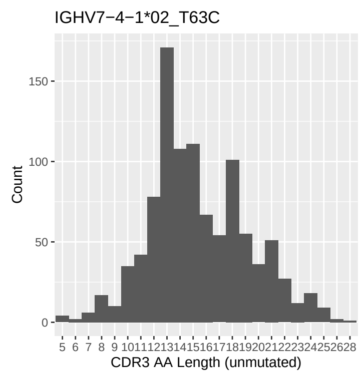

IGHV3−53*01_02_A225G_T267G_G303A_
IGHV3−53*05_T267G_C300T
IGHV3−74*01_A315G
IGHV3−74*01_T279C_A315G
IGHV3−74*01_C246T_G247T_T279C_C2
IGHV4−30−2*01_C315G
IGHV4−30−2*01_C300T_C315G
IGHV4−30−2*01_G245C_C300T_C315G
IGHV4−30−4*01_C315G
IGHV4−31*03_T249C_T285C_C300T
IGHV4−34*01_T300C
IGHV4−38−2*01_A291G_C300T
Count
CDR3 AA Length (unmutated)


IGHV4−NL1*01_C300T
Count
0
5
10
15
20
25
8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 26
CDR3 AA Length (unmutated)

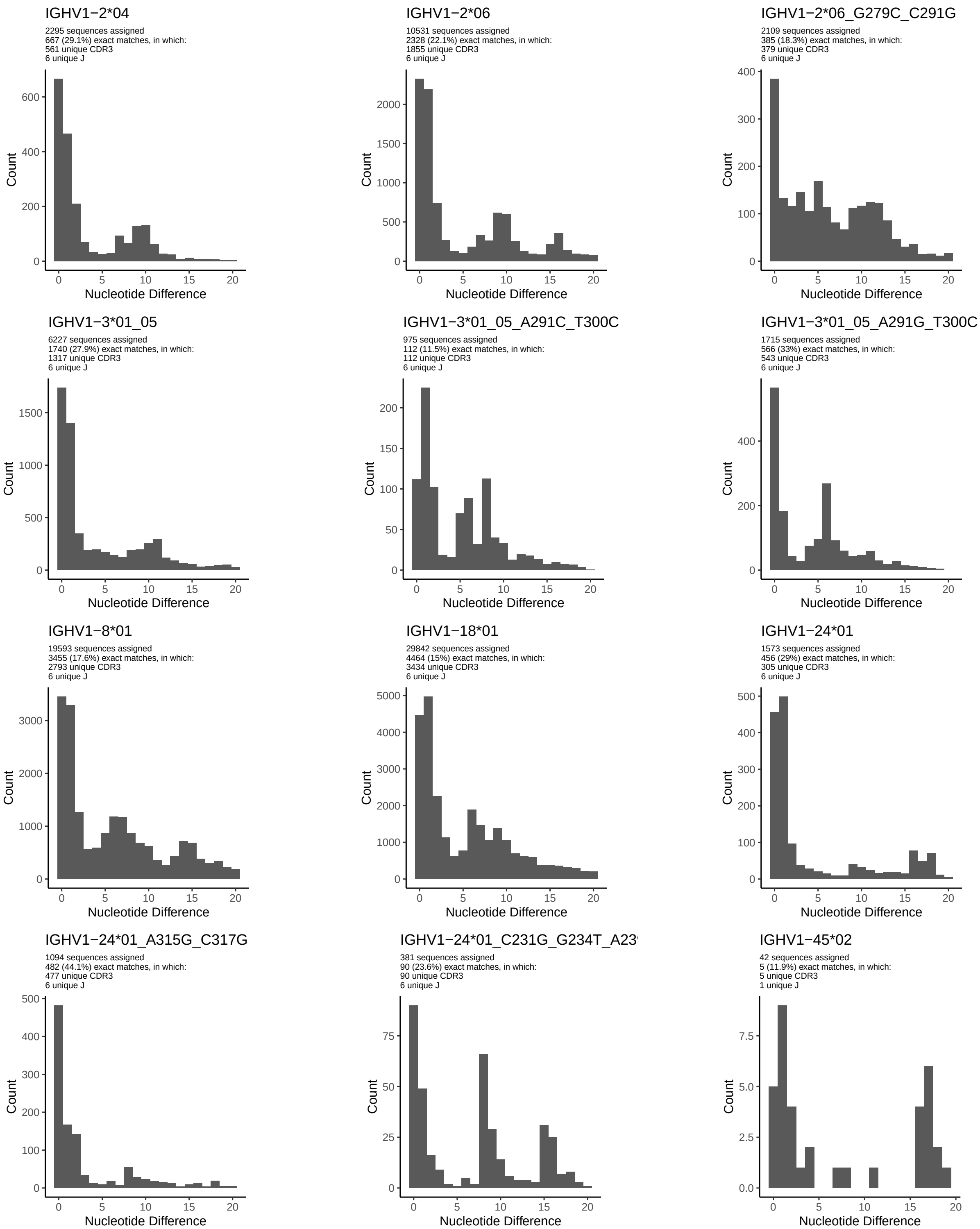

IGHV4−NL1*01_A237C_C300T
Count
0
5
10
15
5 8 9 10 11 12 13 14 15 16 17 18 19 20 21
CDR3 AA Length (unmutated)

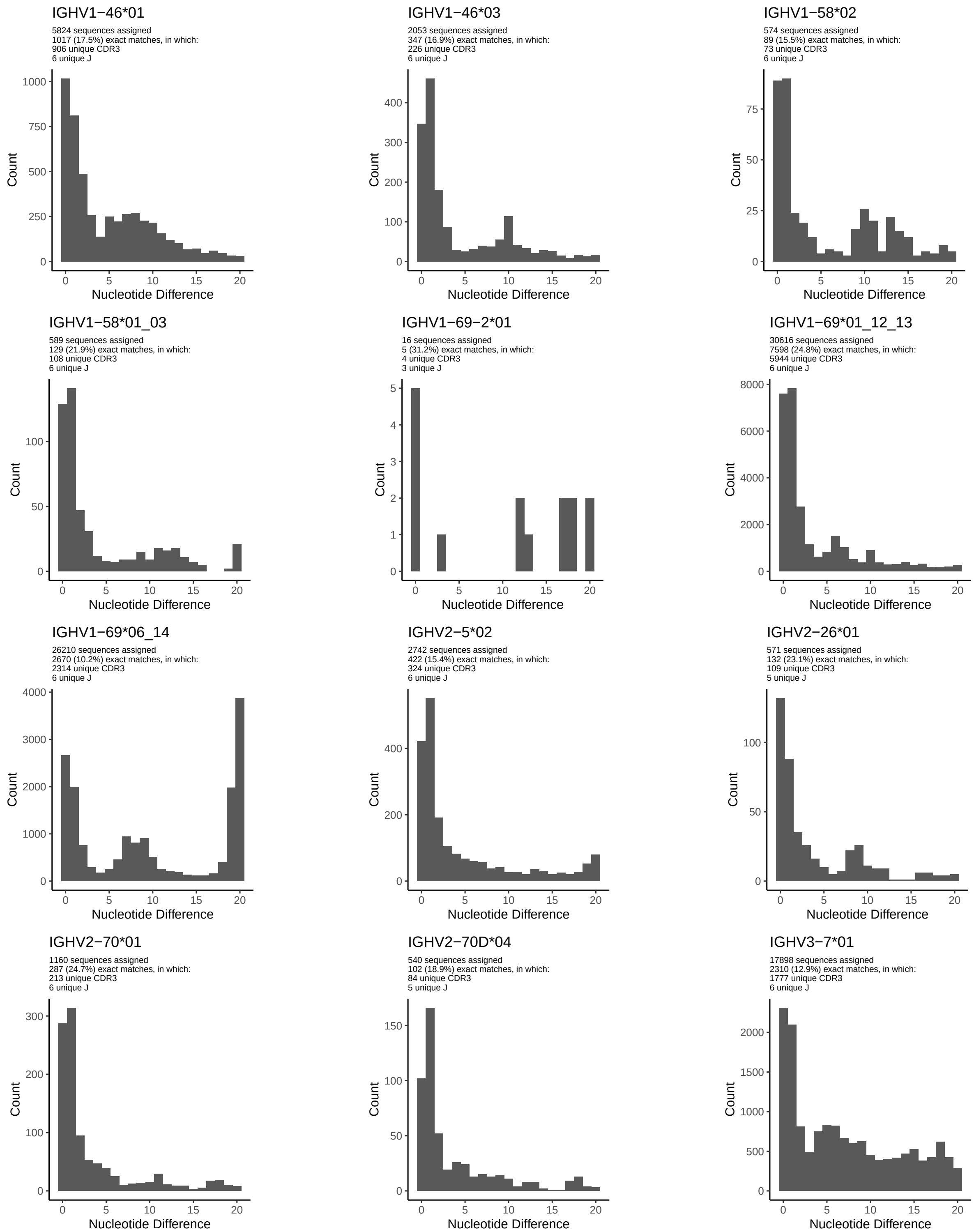

IGHV4−NL1*01_C285T_G315C
Count
0
10
20
30
6 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
CDR3 AA Length (unmutated)

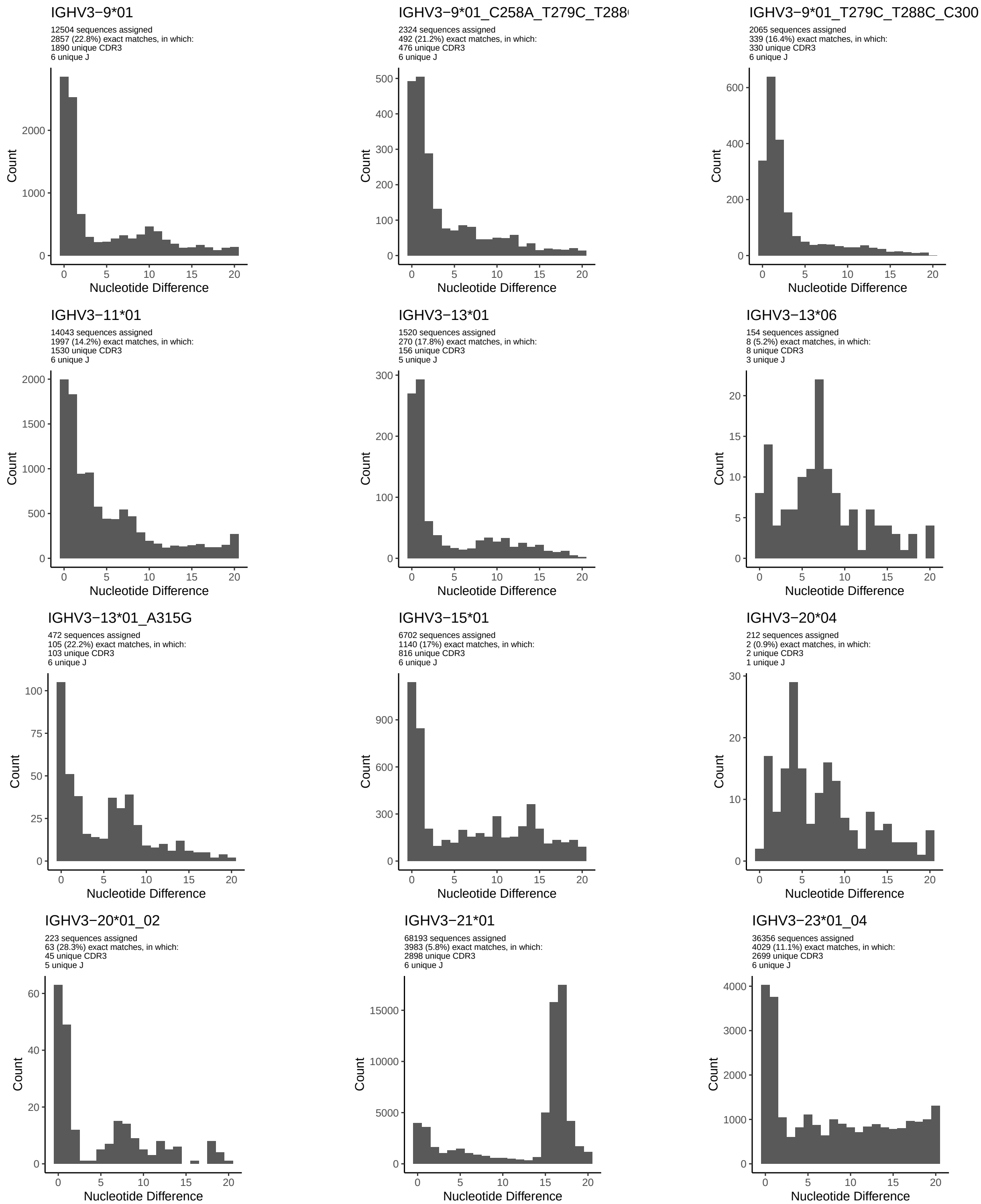

IGHV6−1*01_A315G
Count
0.0
2.5
5.0
7.5
10.0
7 9 10 11 12 13 14 15 16 17 18 20 21 25
CDR3 AA Length (unmutated)

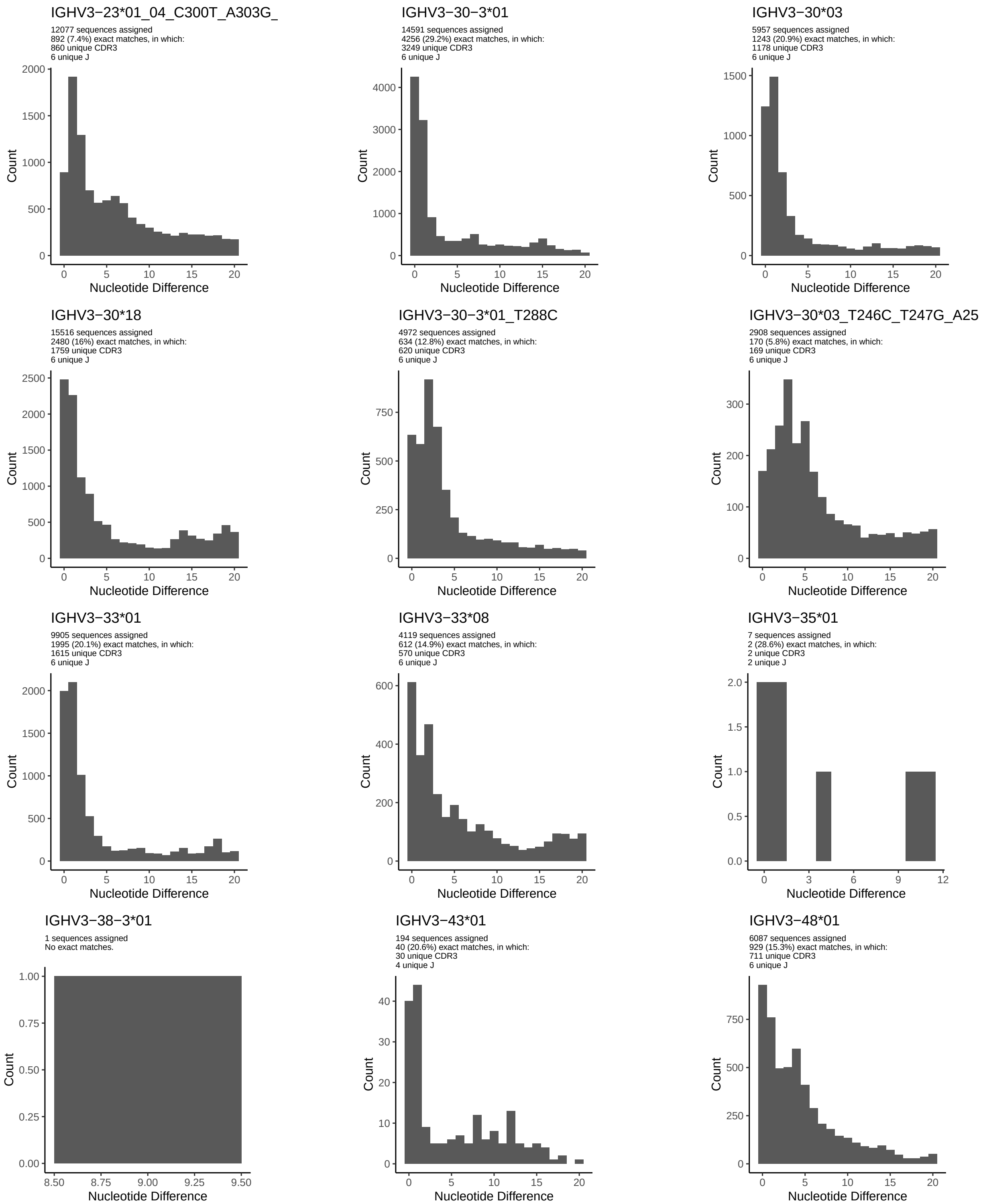

IGHV7−4−1*02_T63C
Count
0
50
100
150
5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 28
CDR3 AA Length (unmutated)

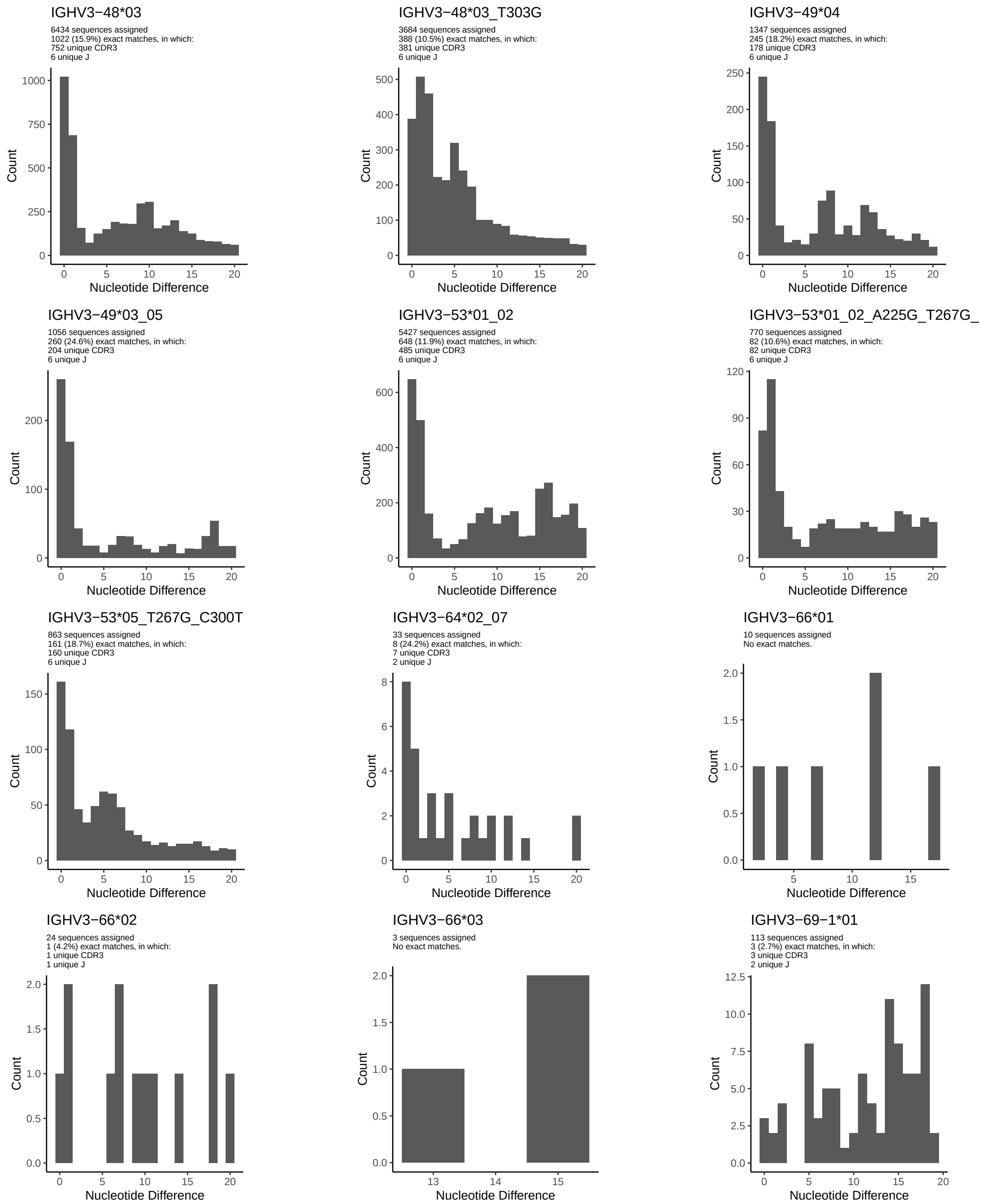
2 Variation from germline, in assignments to each allele

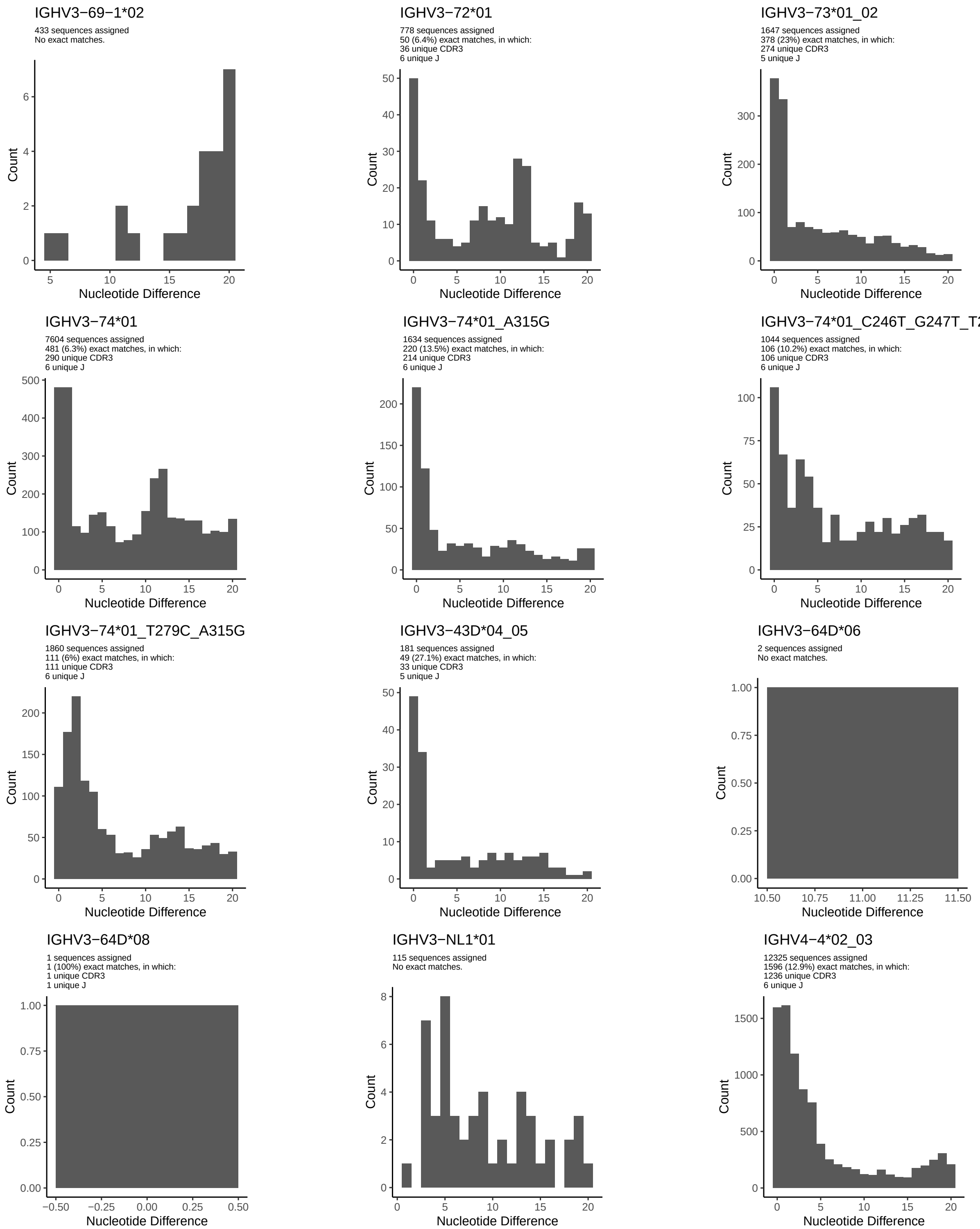


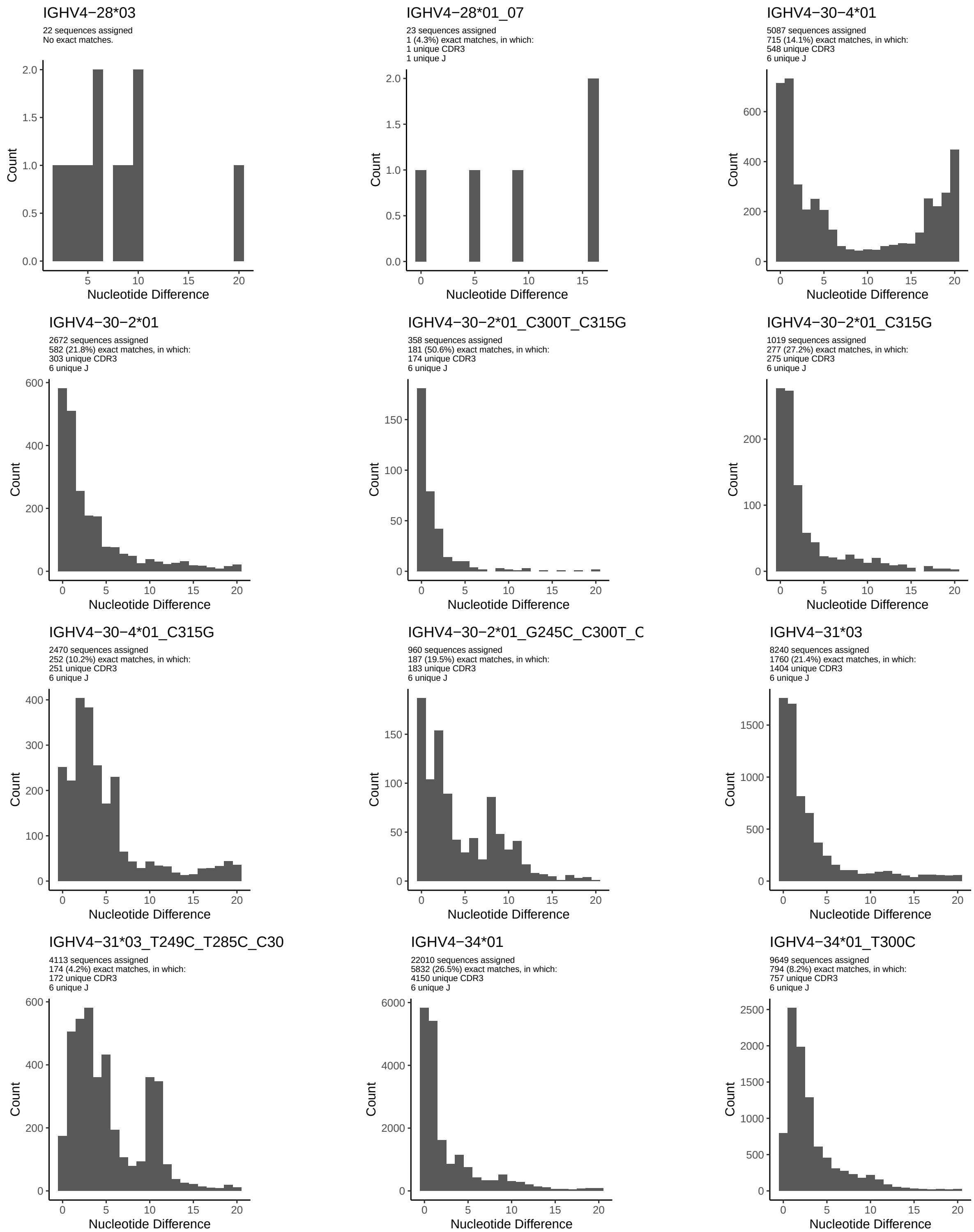

IGHV1−46*01
5824 sequences assigned
1017 (17.5%) exact matches, in which:
906 unique CDR3
6 unique J
IGHV1−46*03
2053 sequences assigned
347 (16.9%) exact matches, in which:
226 unique CDR3
6 unique J
IGHV1−58*02
574 sequences assigned
89 (15.5%) exact matches, in which:
73 unique CDR3
6 unique J
IGHV1−58*01_03
589 sequences assigned
129 (21.9%) exact matches, in which:
108 unique CDR3
6 unique J
IGHV1−69−2*01
16 sequences assigned
5 (31.2%) exact matches, in which:
4 unique CDR3
3 unique J
IGHV1−69*01_12_13
30616 sequences assigned
7598 (24.8%) exact matches, in which:
5944 unique CDR3
6 unique J
IGHV1−69*06_14
26210 sequences assigned
2670 (10.2%) exact matches, in which:
2314 unique CDR3
6 unique J
IGHV2−5*02
2742 sequences assigned
422 (15.4%) exact matches, in which:
324 unique CDR3
6 unique J
IGHV2−26*01
571 sequences assigned
132 (23.1%) exact matches, in which:
109 unique CDR3
5 unique J
IGHV2−70*01
1160 sequences assigned
287 (24.7%) exact matches, in which:
213 unique CDR3
6 unique J
IGHV2−70D*04
540 sequences assigned
102 (18.9%) exact matches, in which:
84 unique CDR3
5 unique J
IGHV3−7*01
17898 sequences assigned
2310 (12.9%) exact matches, in which:
1777 unique CDR3
6 unique J
Count
Nucleotide Difference

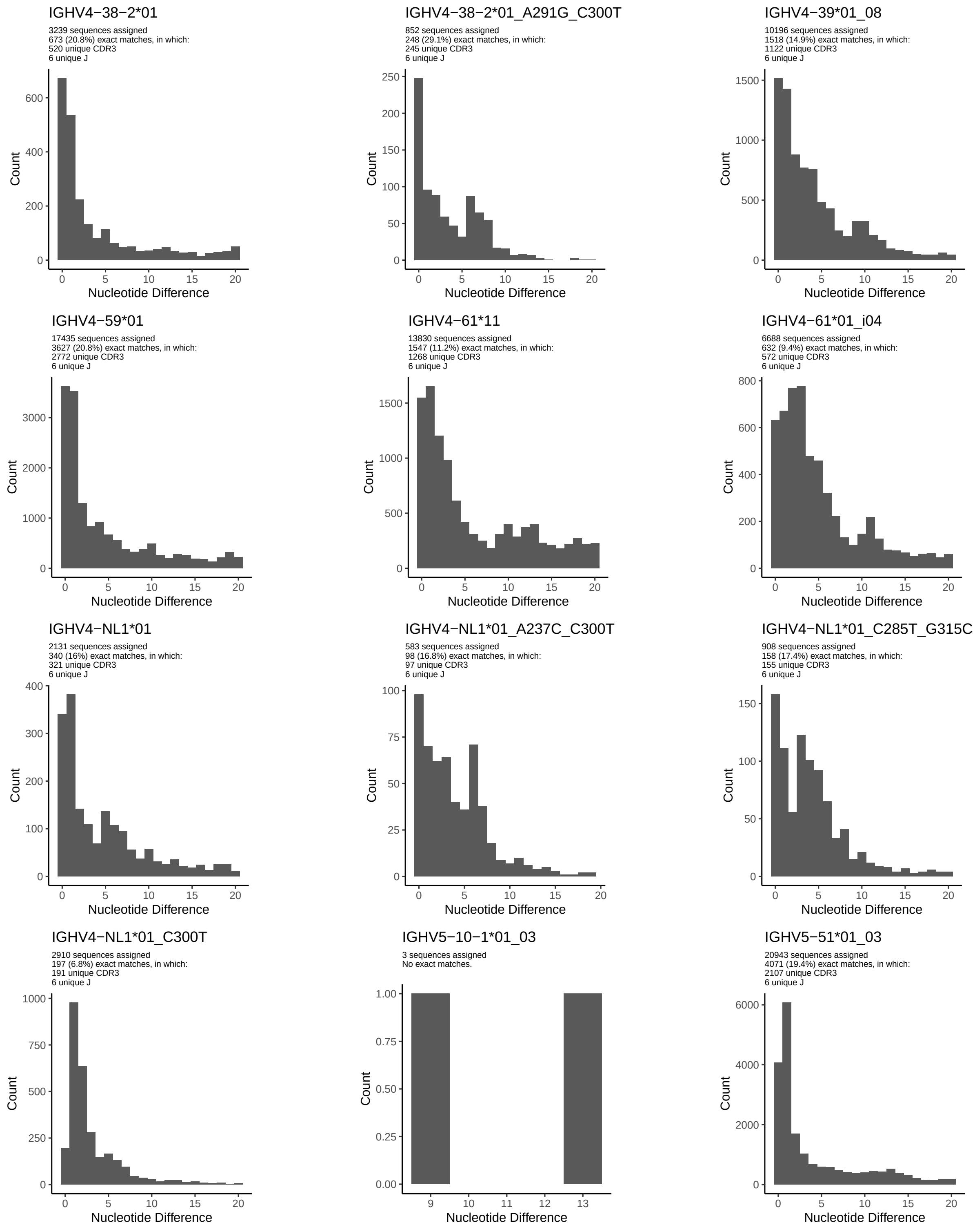

IGHV3−9*01
12504 sequences assigned
2857 (22.8%) exact matches, in which:
1890 unique CDR3
6 unique J
IGHV3−9*01_C258A_T279C_T288
2324 sequences assigned
492 (21.2%) exact matches, in which:
476 unique CDR3
6 unique J
IGHV3−9*01_T279C_T288C_C300
2065 sequences assigned
339 (16.4%) exact matches, in which:
330 unique CDR3
6 unique J
IGHV3−11*01
14043 sequences assigned
1997 (14.2%) exact matches, in which:
1530 unique CDR3
6 unique J
IGHV3−13*01
1520 sequences assigned
270 (17.8%) exact matches, in which:
156 unique CDR3
5 unique J
IGHV3−13*06
154 sequences assigned
8 (5.2%) exact matches, in which:
8 unique CDR3
3 unique J
IGHV3−13*01_A315G
472 sequences assigned
105 (22.2%) exact matches, in which:
103 unique CDR3
6 unique J
IGHV3−15*01
6702 sequences assigned
1140 (17%) exact matches, in which:
816 unique CDR3
6 unique J
IGHV3−20*04
212 sequences assigned
2 (0.9%) exact matches, in which:
2 unique CDR3
1 unique J
IGHV3−20*01_02
223 sequences assigned
63 (28.3%) exact matches, in which:
45 unique CDR3
5 unique J
IGHV3−21*01
68193 sequences assigned
3983 (5.8%) exact matches, in which:
2898 unique CDR3
6 unique J
IGHV3−23*01_04
36356 sequences assigned
4029 (11.1%) exact matches, in which:
2699 unique CDR3
6 unique J
Count
Nucleotide Difference

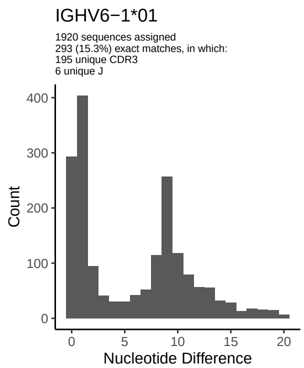

IGHV3−23*01_04_C300T_A303G_
12077 sequences assigned
892 (7.4%) exact matches, in which:
860 unique CDR3
6 unique J
IGHV3−30−3*01
14591 sequences assigned
4256 (29.2%) exact matches, in which:
3249 unique CDR3
6 unique J
IGHV3−30*03
5957 sequences assigned
1243 (20.9%) exact matches, in which:
1178 unique CDR3
6 unique J
IGHV3−30*18
15516 sequences assigned
2480 (16%) exact matches, in which:
1759 unique CDR3
6 unique J
IGHV3−30−3*01_T288C
4972 sequences assigned
634 (12.8%) exact matches, in which:
620 unique CDR3
6 unique J
IGHV3−30*03_T246C_T247G_A25
2908 sequences assigned
170 (5.8%) exact matches, in which:
169 unique CDR3
6 unique J
IGHV3−33*01
9905 sequences assigned
1995 (20.1%) exact matches, in which:
1615 unique CDR3
6 unique J
IGHV3−33*08
4119 sequences assigned
612 (14.9%) exact matches, in which:
570 unique CDR3
6 unique J
IGHV3−35*01
7 sequences assigned
2 (28.6%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV3−38−3*01
1 sequences assigned
No exact matches.
IGHV3−43*01
194 sequences assigned
40 (20.6%) exact matches, in which:
30 unique CDR3
4 unique J
IGHV3−48*01
6087 sequences assigned
929 (15.3%) exact matches, in which:
711 unique CDR3
6 unique J
Count
Nucleotide Difference

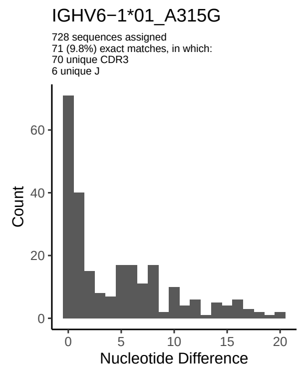

IGHV3−48*03
6434 sequences assigned
1022 (15.9%) exact matches, in which:
752 unique CDR3
6 unique J
IGHV3−48*03_T303G
3684 sequences assigned
388 (10.5%) exact matches, in which:
381 unique CDR3
6 unique J
IGHV3−49*04
1347 sequences assigned
245 (18.2%) exact matches, in which:
178 unique CDR3
6 unique J
IGHV3−49*03_05
1056 sequences assigned
260 (24.6%) exact matches, in which:
204 unique CDR3
6 unique J
IGHV3−53*01_02
5427 sequences assigned
648 (11.9%) exact matches, in which:
485 unique CDR3
6 unique J
IGHV3−53*01_02_A225G_T267G_
770 sequences assigned
82 (10.6%) exact matches, in which:
82 unique CDR3
6 unique J
IGHV3−53*05_T267G_C300T
863 sequences assigned
161 (18.7%) exact matches, in which:
160 unique CDR3
6 unique J
IGHV3−64*02_07
33 sequences assigned
8 (24.2%) exact matches, in which:
7 unique CDR3
2 unique J
IGHV3−66*01
10 sequences assigned
No exact matches.
IGHV3−66*02
24 sequences assigned
1 (4.2%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−66*03
3 sequences assigned
No exact matches.
IGHV3−69−1*01
113 sequences assigned
3 (2.7%) exact matches, in which:
3 unique CDR3
2 unique J
Count
Nucleotide Difference

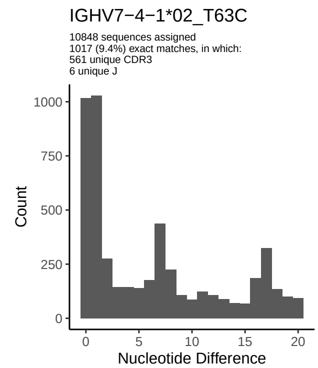

IGHV3−69−1*02
433 sequences assigned
No exact matches.
IGHV3−72*01
778 sequences assigned
50 (6.4%) exact matches, in which:
36 unique CDR3
6 unique J
IGHV3−73*01_02
1647 sequences assigned
378 (23%) exact matches, in which:
274 unique CDR3
5 unique J
IGHV3−74*01
7604 sequences assigned
481 (6.3%) exact matches, in which:
290 unique CDR3
6 unique J
IGHV3−74*01_A315G
1634 sequences assigned
220 (13.5%) exact matches, in which:
214 unique CDR3
6 unique J
IGHV3−74*01_C246T_G247T_T27
1044 sequences assigned
106 (10.2%) exact matches, in which:
106 unique CDR3
6 unique J
IGHV3−74*01_T279C_A315G
1860 sequences assigned
111 (6%) exact matches, in which:
111 unique CDR3
6 unique J
IGHV3−43D*04_05
181 sequences assigned
49 (27.1%) exact matches, in which:
33 unique CDR3
5 unique J
IGHV3−64D*06
2 sequences assigned
No exact matches.
IGHV3−64D*08
1 sequences assigned
1 (100%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−NL1*01
115 sequences assigned
No exact matches.
IGHV4−4*02_03
12325 sequences assigned
1596 (12.9%) exact matches, in which:
1236 unique CDR3
6 unique J
Count
Nucleotide Difference

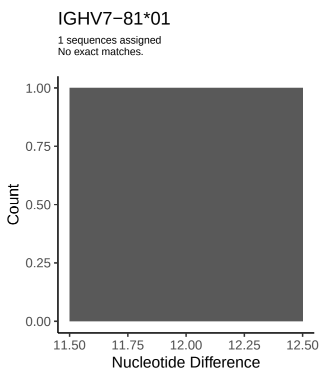

IGHV4−28*03
22 sequences assigned
No exact matches.
IGHV4−28*01_07
23 sequences assigned
1 (4.3%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV4−30−4*01
5087 sequences assigned
715 (14.1%) exact matches, in which:
548 unique CDR3
6 unique J
IGHV4−30−2*01
2672 sequences assigned
582 (21.8%) exact matches, in which:
303 unique CDR3
6 unique J
IGHV4−30−2*01_C300T_C315G
358 sequences assigned
181 (50.6%) exact matches, in which:
174 unique CDR3
6 unique J
IGHV4−30−2*01_C315G
1019 sequences assigned
277 (27.2%) exact matches, in which:
275 unique CDR3
6 unique J
IGHV4−30−4*01_C315G
2470 sequences assigned
252 (10.2%) exact matches, in which:
251 unique CDR3
6 unique J
IGHV4−30−2*01_G245C_C300T_C
960 sequences assigned
187 (19.5%) exact matches, in which:
183 unique CDR3
6 unique J
IGHV4−31*03
8240 sequences assigned
1760 (21.4%) exact matches, in which:
1404 unique CDR3
6 unique J
IGHV4−31*03_T249C_T285C_C30
4113 sequences assigned
174 (4.2%) exact matches, in which:
172 unique CDR3
6 unique J
IGHV4−34*01
22010 sequences assigned
5832 (26.5%) exact matches, in which:
4150 unique CDR3
6 unique J
IGHV4−34*01_T300C
9649 sequences assigned
794 (8.2%) exact matches, in which:
757 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−38−2*01
3239 sequences assigned
673 (20.8%) exact matches, in which:
520 unique CDR3
6 unique J
IGHV4−38−2*01_A291G_C300T
852 sequences assigned
248 (29.1%) exact matches, in which:
245 unique CDR3
6 unique J
IGHV4−39*01_08
10196 sequences assigned
1518 (14.9%) exact matches, in which:
1122 unique CDR3
6 unique J
IGHV4−59*01
17435 sequences assigned
3627 (20.8%) exact matches, in which:
2772 unique CDR3
6 unique J
IGHV4−61*11
13830 sequences assigned
1547 (11.2%) exact matches, in which:
1268 unique CDR3
6 unique J
IGHV4−61*01_i04
6688 sequences assigned
632 (9.4%) exact matches, in which:
572 unique CDR3
6 unique J
IGHV4−NL1*01
2131 sequences assigned
340 (16%) exact matches, in which:
321 unique CDR3
6 unique J
IGHV4−NL1*01_A237C_C300T
583 sequences assigned
98 (16.8%) exact matches, in which:
97 unique CDR3
6 unique J
IGHV4−NL1*01_C285T_G315C
908 sequences assigned
158 (17.4%) exact matches, in which:
155 unique CDR3
6 unique J
IGHV4−NL1*01_C300T
2910 sequences assigned
197 (6.8%) exact matches, in which:
191 unique CDR3
6 unique J
IGHV5−10−1*01_03
3 sequences assigned
No exact matches.
IGHV5−51*01_03
20943 sequences assigned
4071 (19.4%) exact matches, in which:
2107 unique CDR3
6 unique J
Count
Nucleotide Difference

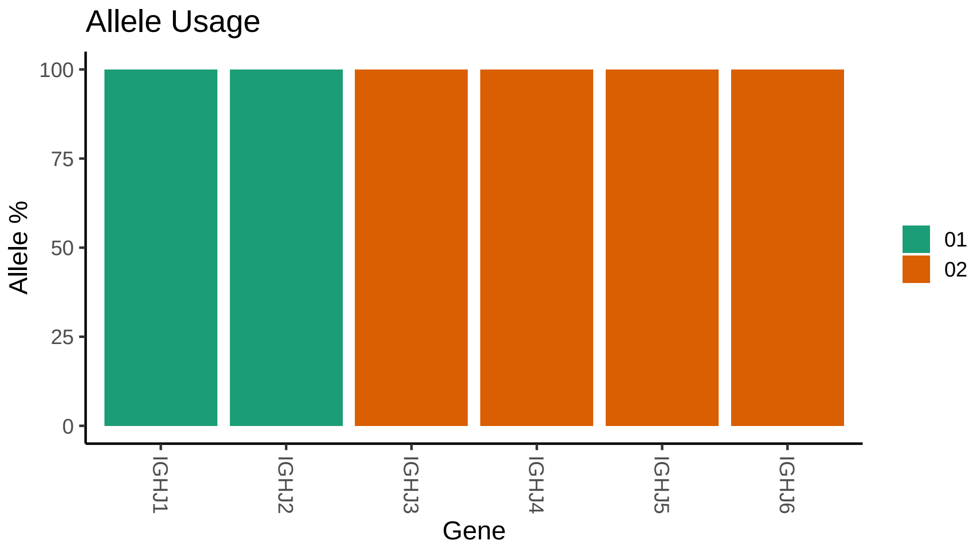

IGHV6−1*01
1920 sequences assigned
293 (15.3%) exact matches, in which:
195 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV6−1*01_A315G
728 sequences assigned
71 (9.8%) exact matches, in which:
70 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV7−4−1*02_T63C
10848 sequences assigned
1017 (9.4%) exact matches, in which:
561 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV7−81*01
1 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /misc/work/jenkins/PRJNA491287/M64 099/M64 099/M64 099/M64 099/7b/cc6733d6fddd92fa89
##
## Germline reference file: /misc/work/jenkins/PRJNA491287/M64 099/M64 099/M64 099/M64 099/7b/cc6733d6fd
##
## Novel allele file: /misc/work/jenkins/PRJNA491287/M64 099/M64 099/M64 099/M64 099/7b/cc6733d6fddd92fa
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 2 06: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 2 06_G279C_C291G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291C_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_C231G_G234T_A239C_A244G_C245A_T249C_A252G_G253A_A254G_A31
##
##
## Unexpected N in reference sequence IGHV3 9 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 9 01_T279C_T288C_C300T_T301G_A315G_A317G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 9 01_C258A_T279C_T288C_C300T_T301G_A315G_A317G: replacing with g
##
##
## Unexpected N in reference sequence IGHV3 13 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 13 01_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T246C_T247G_A256T_G258A_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 3 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 48 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 03_T303G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 53 01_02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 53 01_02_A225G_T267G_G303A_G317A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 66 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 53 05_T267G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_T279C_A315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_C246T_G247T_T279C_C288T_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C300T_C315G: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_G245C_C300T_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 4 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 31 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 31 03_T249C_T285C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 38 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 01_A291G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_A237C_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C285T_G315C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV6 1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV6 1 01_A315G: replacing with gap
```